

A Brief Introduction to Graph Theory

Working Draft

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Part I

Fundamentals

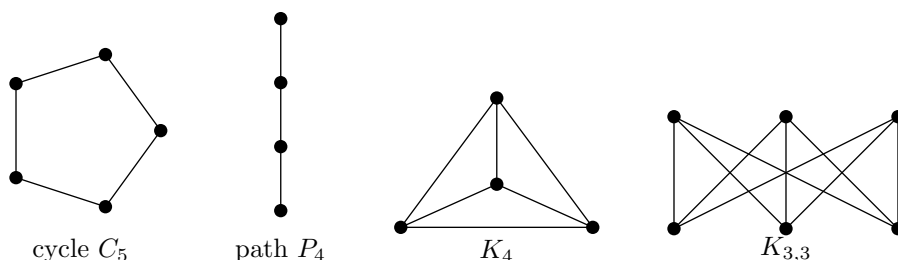
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Chapter 1: Graphs, Subgraphs, Degrees, and Connections

1.1 Graphs and Subgraphs

A **graph** is a set of vertices (singular vertex) and edges, where every edge joins two vertices. A **simple** graph is one without loops or multiple edges; we will assume our graph is simple unless otherwise stated. Two vertices are said to be **neighbors** or **adjacent** if they are joined by an edge. A graph is **connected** if one can get from each vertex to every other vertex by a sequence of edges. The **components** of a graph are the connected pieces.

There are four common graphs. We use P_n to denote the **path** on n vertices, C_n to denote the **cycle** on n vertices, and K_n to denote the **complete graph** on n vertices. We also use $K_{r,s}$ to denote the **complete bipartite graph** constructed by taking a set of r vertices and another set of s vertices, and adding all possible edges between the two sets. The graph P_n has $n - 1$ edges, the graph C_n has n edges, the complete graph K_n has $\binom{n}{2}$ edges, and the complete bipartite graph $K_{r,s}$ has rs edges. Here are some pictures.

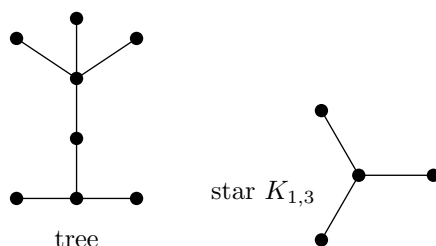


A **subgraph** of a graph has some of the vertices and some of the edges. An **induced subgraph** is one which contains all possible edges for its vertices. A **clique** is a set of vertices such that every pair of vertices are adjacent; equivalently, a subgraph that is complete. An **independent set** is a set of vertices such that no pair of vertices is adjacent.

EXAMPLE. The maximum size of an independent set is: $\lceil n/2 \rceil$ for P_n , $\lfloor n/2 \rfloor$ for C_n , 1 for K_n , and $\max\{r, s\}$ for $K_{r,s}$. The maximum size of a clique is 2

for P_n (unless $n = 1$), for C_n (unless $n = 3$) and for $K_{r,s}$.

A graph without a cycle is called **acyclic** or a **forest**. A **tree** is a connected acyclic graph. Here is a forest of two trees.



1.2 Degrees, Paths, and Eulerian Tours

The **degree** of a vertex v , denoted $d(v)$, is the number of edges incident with it (except loops count twice). A graph is **k -regular** if every vertex has degree k .

FIRST THEOREM OF GRAPH THEORY. *The sum of the degrees of the graph is twice the number of edges.*

PROOF. We use double counting. Let A be the number of ends of edges. Then A is twice the number of edges, since every edge has two ends. On the other hand, every end of an edge contributes one to the degree of that vertex. So summing up the degrees gives the total number of ends of edges. $\square$

It follows that in any graph the number of vertices of odd degree is even.

A **walk** is a sequence of edges. A **trail** is a walk without repeated edges (but can have repeated vertices). A **path** is a trail without repeated vertices. A **cycle** is almost a path except the first and last vertices are the same. A walk is **closed** if the first and last vertices are the same. The **length** of a walk is the number of edges.

There are several interesting properties of trees:

PROPERTIES OF TREES. *For every tree:*
 (a) *The number of edges is one less than the order.*

- (b) Every pair of vertices are joined by a unique path.
- (c) The removal of any edge disconnects the tree.
- (d) The addition of any edge creates a unique cycle.
- (e) If the tree has at least one edge, then it has at least two vertices of degree 1 (called **leaves**).

Indeed, one can characterize trees by such properties. It follows that:

OBSERVATION. *If a graph is connected, then the number of edges is at least one less than the order.*

A graph is **bipartite** if the vertices can be partitioned into two sets (called **partite sets**) such that every edge has an endpoint in both sets. Equivalently, one can color the vertices with two colors, say red and blue, such that both the red vertices and the blue vertices form an independent set. For example, trees and even cycles are bipartite. It can be shown that if a bipartite graph is regular, then the partite sets have equal size.

CHARACTERIZATION OF BIPARTITE GRAPHS. *A graph is bipartite if and only if it contains no odd cycle.*

PROOF. It is straightforward that an odd cycle is not bipartite. Thus every graph containing an odd cycle is not bipartite. To show the other direction, we provide a recipe to color the graph: either the recipe produces a bipartite coloring or it produces an odd cycle. Pick a vertex v and color in rounds: in round 0 color v red, in round 1 color its neighbors blue, in round 2 color their neighbors red, and so on. Note that two vertices that are adjacent are colored in the same round or in consecutive rounds. Suppose at the end there is an edge whose ends x and y have the same color. Then the two vertices were colored in the same round. For each vertex, we can construct a path from it back to v . Suppose the two paths back from x and y first meet up at vertex v' . Then the portion of the paths up to v' together with the edge xy form an odd cycle. $\square$

An **Eulerian tour** is a closed trail that uses every edge. A graph is **Eulerian** if it has an Eulerian tour.

CHARACTERIZATION OF EULERIAN GRAPHS. *A graph without isolated vertices has an Eulerian tour if and only if it is connected and every vertex has even degree.*

PROOF. (1) Assume we have an Eulerian tour. Every vertex other than the start must have even degree, since every time we visit a vertex we use up two edges incident with it. Similarly, the start vertex has even degree: apart from the first and last edge of the tour we use up the edges in pairs. It follows that every degree is even.

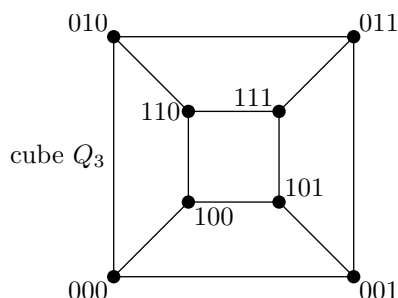
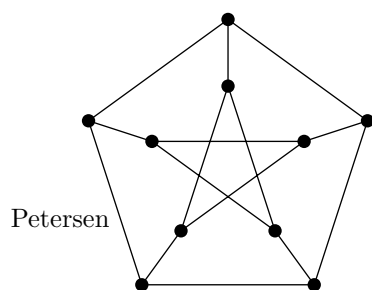
(2) Define a ***tour decomposition*** as a collection of tours that use up all the edges of the graph. Assume the graph has every vertex of even degree. Then we claim there is a tour decomposition.

One can construct a tour decomposition in a simple fashion: start tracing out a tour until one gets stuck. The only place one can get stuck is back where one started, since for every other vertex there is always another edge to leave it. Since a tour uses an even degree at each vertex, once we remove the first tour, what remains must have even degree throughout. Thus, one can repeat the process, thereby using up all the edges.

But what we want is an Eulerian tour. So take the tour decomposition $\mathcal{T}$ with the minimum number of tours. And suppose there is more than one tour in $\mathcal{T}$.

Then, because the original graph is connected, there must be somewhere two tours that share a vertex, call it p . We can re-organize things such that p is the start and finish of both tours. But then it is easy to merge the two tours, by continuing on the second after finishing the first. Hence we get a tour decomposition with fewer tours, a contradiction. That is, in fact, $\mathcal{T}$ contains an Eulerian tour. $\square$

We finish this section by introducing two common graphs. The Petersen graph has 10 vertices, is 3-regular, and the shortest cycle has length 5. The k -cube or hypercube Q_k is the graph whose vertices are all k -tuples with 0 and 1, and edges if the two k -tuples differ in exactly one position. It is bipartite, has 2^k vertices and is k -regular. The hypercube also has a recursive definition, being built up from smaller hypercubes. Here are the pictures.



1.3 Distance, Degrees, and Decompositions

The **degree sequence** of a graph is the sorted list of the degrees of its vertices. Any degree sequence is possible provided the sum of degrees is even. However, if one needs a simple graph, then there are several characterizations, the most famous of which is by Havel–Hakimi.

A **decomposition** of a graph is a partition of the edges (each edge is in exactly one part of the partition). The concept of Eulerian tour can be generalized to characterize graphs that can be decomposed into k trails.

Two graphs are **isomorphic** if there is a bijection between the vertex sets that preserves the edges. The **complement** of a simple graph has the same vertex set but the missing edges. A graph is **self-complementary** if it is isomorphic to its complement (e.g. P_4 or C_5). An exercise is to show that the order must be congruent to 0 or 1 modulo 4.

The **distance** from u to v is the length of the shortest path from u to v . The **eccentricity** of a vertex is the maximum distance of a vertex from it. The **radius** is the minimum eccentricity and the **center** is those vertices of minimum eccentricity. The **diameter** is the maximum eccentricity, or equivalently, the maximum distance between vertices. The **girth** is the shortest cycle length.

DISTANCE BOUNDS. *For any connected graph,*
 (a) $\text{rad} \leq \text{diam} \leq 2 \text{rad}$.
 (b) *If the graph is not a tree then* $\text{girth} \leq 2 \text{rad} + 1$

PROOF. (a) Consider any two vertices x and y , and let v be a vertex in the center (that is, having minimum eccentricity). Then there is a walk from x to y

via v ; thus the distance from x to y is at most 2rad .

(b) Suppose v is a vertex in the center. Consider a tree subgraph T formed by taking for each vertex a shortest path from it to v . Since the whole graph is not a tree, there is some edge not in T ; that together with T creates a cycle of length at most $2\text{rad}+1$. $\square$

The **Wiener index** of a graph is the sum of the distances between every pair of vertices. (If divide by $\binom{n}{2}$ then get the **average distance**.) For trees of order n , the Wiener index is maximum for a path and minimum for a star.

Chapter 2: Connectivity, Spanning Trees, and Directed Graphs

2.1 Connectivity

A **cut-vertex** is a vertex whose removal disconnects its component. A **bridge** is an edge whose removal disconnects its component: an edge is a bridge if and only if it belongs to no cycle. A **block** is a maximal connected subgraph that has no cut-vertex.

A **cut-set** is a set of vertices S such that $G - S$ has more than one component. The **connectivity** $\kappa(G)$ of graph G is the size of the smallest cut-set, except that we define $\kappa(K_n) = n - 1$. The **edge-connectivity** $\kappa'(G)$ is the smallest cardinality of a disconnecting set of edges, except that we define $\kappa'(K_1) = 0$.

For a graph G , the notation $\delta(G)$ means the minimum degree and $\Delta(G)$ means the maximum degree. Fundamental inequality is:

INEQUALITY CHAIN. *For any graph G we have $\kappa(G) \leq \kappa'(G) \leq \delta(G)$.*

PROOF. To show that $\kappa' \leq \delta$, note that removing all edges incident with a vertex of minimum degree disconnects the graph. To show that $\kappa \leq \kappa'$, we observe that both parameters are 0 if the graph is disconnected. So assume the graph is connected, and let X be a minimum disconnecting set of edges, say containing edge e . If one were to remove all edges of X except e , what remains is connected, by the minimality of X . So, for each such edge, consider one of the ends that is not an end of e , and remove that from the graph. If the result is disconnected, or there are only two vertices remaining, then the bound is proved. So assume otherwise. Then e is a bridge, and removing one end of e disconnects the graph. $\square$

It can be shown that if graph is 3-regular, then $\kappa = \kappa'$.

EXAMPLE. The hypercube Q_k has connectivity k . Think of Q_k as two copies of Q_{k-1} where each pair of corresponding vertices are joined by an edge. We prove by induction, that removing $k - 1$ vertices does not disconnect Q_k . If we remove no vertex from one of the copies of Q_{k-1} , then that copy retains

intact and every remaining vertex is connected to that copy. If we remove some vertices from each copy of Q_{k-1} totaling $k-1$, then by the induction hypothesis both copies remain connected, and since $k-1 < 2^{k-1}$, some edge joining the two copies remains.

2.2 Spanning Trees

A **spanning subgraph** is one with all the vertices and only some of the edges. One can transform any spanning tree into any other spanning tree by changing one edge at a time (that is, a **greedy algorithm**.) Taking an end-vertex of a spanning tree shows that every nontrivial graph has a vertex that is not a cut-vertex.

A famous result is the following:

CAYLEY'S FORMULA. *The labeled K_n has n^{n-2} different spanning trees.*

PROOF SKETCH. The proof idea is to prove a bijection between the set of spanning trees on the one hand, and the set of sequences of length $n-2$ whose entries are drawn from $\{1, \dots, n\}$ on the other hand. There are clearly n^{n-2} such sequences.

We label the vertices of K_n with 1 up to n , and given a spanning tree create what is called the Prüfer code. At each stage, delete the smallest-numbered remaining leaf and write down its neighbor. Stop when two vertices are left.

The surprising fact is that this process is reversible. It can be argued that different trees have different codes and that given such a sequence one can build a spanning tree with that Prüfer code. $\square$

The notation $G-e$ means the graph obtained by deleting edge e . The notation $G \cdot e$ means the graph obtained by contracting edge e , meaning merge its two ends into one vertex; note that the result might not be a simple graph.

CONTRACTION-DELETION. *If τ is the number of spanning trees then $\tau(G) = \tau(G-e) + \tau(G \cdot e)$ for any edge e .*

PROOF. Consider a spanning tree T of graph G . If T does not contain edge e , then T is a spanning tree of $G - e$ and conversely. If T does contain edge e , then $T \cdot e$ is a spanning tree of $G \cdot e$, and conversely, every spanning tree of $G \cdot e$ becomes a spanning tree of G when e is “uncontracted”. $\square$

The **adjacency matrix** is the square matrix where the entry in row i column j is 1 if there is an edge between vertices i and j , and 0 otherwise. The **Laplacian matrix** is $D - A$ where D is the diagonal matrix with the degrees and A is the adjacency matrix. We omit the proof of:

MATRIX TREE THEOREM. *The number of spanning trees is given by the absolute value of any cofactor (determinant of matrix with one row and column deleted) of the Laplacian.*

Graceful labeling of tree: A labeling of the vertices with 0 up to $n - 1$ so that vertices have distinct labels and the edge-differences are distinct. A famous unsolved conjecture is that every tree has a graceful labeling.

In graphs with nonnegative edge-weights, Kruskal’s algorithm finds a **minimum-weight spanning tree**. The algorithm says repeatedly take the cheapest edge that does not create a cycle. (Proof of validity of Kruskal is by switches: comparing the alleged minimum spanning tree with one from Kruskal.)

A **breadth-first-search tree** is a spanning tree created by rooting at a vertex, taking its neighbors, then their neighbors, and so on. This can be generalized to calculate the distance from a vertex to all other vertices, in Dijkstra’s algorithm.

2.3 Some Extremal Results

Many questions in graph theory ask about the minimum or maximum of a quantity (such as edges) for graphs with or without some property. An **extremal** graph is an example with the minimum or maximum.

MANVEL. *The maximum number of edges in a triangle-free simple graph of order n is $\lfloor n^2/4 \rfloor$.*

PROOF. Start with a vertex v of maximum degree Δ . Let $N(v)$ be its neighborhood. Because the graph has no triangle, the set $N(v)$ is independent. This

means that every edge of the graph has one or two ends outside $N(v)$. So when we sum up the degrees of the vertices outside $N(v)$, this counts all edges (and some of them twice). There are $n - \Delta$ vertices outside $N(v)$ and each has degree at most Δ . So we get that the total number of edges is at most $\Delta(n - \Delta)$. Then by calculus this is maximized at $\Delta = n/2$. The extremal graph is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$. $\square$

BIPARTITE SUBGRAPHS. *The complete graph K_n can be:*

(a) *decomposed into $\lceil \log_2 n \rceil$ bipartite subgraphs.*

(b) *decomposed into $n - 1$ complete bipartite subgraphs.*

And these numbers are best possible.

PROOF SKETCH. We show only the existence here. (a) Assume n is a power of 2. For the first subgraph, partition the vertex set of K_n into two equal sets and take the complete bipartite subgraph with those as partite sets. What remains is two disjoint copies of $K_{n/2}$ and one can use the same color scheme on both copies.

(b) Color all the edges incident with the first vertex with the same color. What remains is K_{n-1} and one can repeat. $\square$

Decomposition results can also be expressed as union results. The ***union*** of two graphs is constructed by unioning their vertex sets and edge sets (throwing away duplicates). The ***disjoint union*** of two graphs is the union assuming their vertex sets are disjoint (and result is guaranteed to be disconnected).

TREE SUBGRAPHS. *If a graph has minimum degree at least k , then it contains as subgraphs all trees with at most k edges*

PROOF. Build each desired tree one vertex at a time. $\square$

2.4 Directed Graphs

A ***directed graph*** or ***digraph*** has directed edges (sometimes called arcs). Each arc has a head and a tail.

A path in a digraph must respect the arrows. A digraph is strongly connected if there is path from every vertex to every other vertex. The strong components

are its maximal strongly connected subgraphs. Subgraph, isomorphism, decomposition, union is similar to before. Examples include: Markov chains, finite automata, and game graphs

The *in-degree* and *out-degree* of a vertex are the numbers of edges going in or going out of the vertex. If the out-degree is positive for all vertices, then the digraph has a cycle; similarly with in-degree always positive. It can be shown that a connected digraph is Eulerian if and only if the in-degree equals out-degree at every vertex.

An *orientation* of a graph means choosing one orientation of each edge. A *tournament* is an oriented K_n . We define a *king* as a vertex that can reach all other vertices with a path of length at most 2.

KINGS. *Every tournament has a king.*

PROOF. We claim that every vertex v of maximum out-degree is a king. For, consider any other vertex w . If there is an arc from v to w , then v reaches w in one. So assume not; that is, there is an arc from w to v . Since the out-degree of w is at most that of v , there must be some vertex x such that x is an out-neighbor of v but not of w . This means v reaches w in two going $v \rightarrow x \rightarrow w$. $\square$

Chapter 3: Matchings

3.1 Matchings and Perfect Matchings

A *matching* in a graph is a set of edges no two of which are adjacent. A vertex is *saturated* if it is the end of an edge in the matching, and a *perfect matching* (sometimes called a 1-factor) saturates all the vertices.

EXAMPLE. Consider counting the perfect matchings of the complete graph K_{2n} . One way to proceed is to choose an ordering of the vertices—in $(2n)!$ ways—and build a perfect matching by pairing first and second, third and fourth, and so on. But since which vertex is first and which is second doesn't matter, and the order of the edges doesn't matter, any particular matching is built in $n!2^n$ different ways. Another way to proceed is to choose the partner for the smallest numbered vertex, in $2n - 1$ ways. Then the partner for the smallest unmatched vertex can be chosen in $2n - 3$ ways. And so on. Thus the number of perfect matchings is

$$\frac{2n}{2^n n!} = (2n - 1)(2n - 3) \cdots 1$$

(This quantity is sometime written as $(2n)!!$ with the double factorial.)

In general we care about the maximum size of a matching. Note that a maximal matching is not the same as a maximum matching: the middle edge of P_4 is a maximal matching but not maximum. There is a famous characterization of when a matching M is maximum. Given a matching M , an *M -alternating* path is one where the edges alternate between in and out of M . An *M -augmenting* path is a nontrivial M -alternating path that starts and ends at an M -unsaturated vertex.

BERGE. A matching M is maximum if and only if there is no M -augmenting path.

PROOF. Assume first there is an M -augmenting path. Then one can flip membership of the edges on the path, say from out-in-out-in-out to in-out-in-out-in, and so increase the size of the matching by 1. That is, M is not maximum.

Assume second that M is not maximum. For two matchings (or graphs in general) define the ***symmetric difference*** as the subgraph with the edges that are in precisely one of the graphs. The key observation is that, if M_1 and M_2 are matchings, then their symmetric difference consists of paths and even cycles. (It is immediate that the symmetric difference has maximum degree 2 and so consists of paths and cycles; but the edges alternate being in M_1 and M_2 so only even cycles are possible.)

Now, let M^* be a maximum matching and consider the symmetric difference S of M and M^* . Since M^* has more edges than M in G , this also holds in S . So there is a component of S where M^* is in the majority. This must be an odd path, starting and finishing with an M^* edge, and by the way S is constructed, the ends of the path are not M -saturated. That is, the path is M -augmenting. $\square$

Berge's theorem gives an algorithm to find a maximum matching in a bipartite graph. Starting at an unmatched vertex, one determines the set of vertices reachable by M -alternating paths, and thus either finds an M -augmenting path or proves that none exists.

Another characterization of when a graph has a perfect matching is based on odd components (meaning odd order).

TUTTE'S 1-FACTOR THEOREM. *A graph G has a perfect matching if and only if $o(G - S) \leq |S|$ for all sets of vertices S , where $o(G - S)$ is the number of odd components of $G - S$.*

PROOF SKETCH. For each odd component of $G - S$, one of its vertices needs to be matched with a vertex of S . Thus if $o(G - S) > |S|$ then a perfect matching is not possible. We omit the proof of the reverse direction. $\square$

The following result can be deduced from the previous:

PETERSEN'S THEOREM. *A 3-regular graph without a bridge has a perfect matching.*

3.2 Vertex Covers and Matching in Bipartite Graphs

For a set of vertices X , we let $N(X)$ denote the *neighborhood* of X ; that is, all vertices that have a neighbor in X .

HALL'S THEOREM. *In a bipartite graph with one side X : there is a matching that saturates X if and only if $|N(S)| \geq |S|$ for all $S \subseteq X$.*

PROOF. The condition is obviously necessary, as a saturating matching provides a distinct neighbor for every vertex in X . So we focus on the proof of sufficiency.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. Suppose the maximum matching M does not saturate vertex x_0 in X . Then the condition means x_0 has a neighbor, say y_0 . Since the path x_0y_0 is not augmenting, it must be that y_0 is saturated, say with edge to x_1 . Since $|N(\{x_0, x_1\})| \geq 2$, there is a vertex y_1 that is adjacent to one or both of x_0 and x_1 . In either case there is an M -alternating path from x_0 to y_1 ; thus y_1 must be saturated. And so on. We build a set $y_0, y_1, \dots$ each vertex of which is reachable from x_0 by an M -alternating path, and when this number exceeds the number of edges in M , we have an M -augmenting path. $\square$

An immediate corollary is:

REGULAR BIPARTITE. *A regular bipartite graph can be decomposed into perfect matchings.*

PROOF. Let G be a k -regular bipartite graph (with $k > 0$). We observed earlier that the two partite sets have the same size. We claim that it satisfies the condition in Hall's Theorem. Let A be the number of edges between S and $N(S)$. Then $k|S| = A \leq k|N(S)|$, by the degrees. It follows that G has a matching saturating X . Finally note, that such a matching is a perfect matching, and when we remove it, what remains is a $(k - 1)$ -regular bipartite graph. $\square$

A **vertex cover** is a set of vertices intersecting all edges. It is immediate that a set is a vertex cover if and only if its complement is an independent set. An **edge cover** is a set of edges saturating all vertices. Notation: $\alpha(G)$ is the maximum size of an independent set of vertices, $\beta(G)$ is the minimum size of a

vertex cover, $\alpha'(G)$ is the maximum size of a matching, and $\beta'(G)$ is minimum size of edge cover. (Note that we use the primes for the edge versions.)

KÖNIG–EGERVÁRY. *If G is a bipartite graph then $\alpha'(G) = \beta(G)$.*

PROOF SKETCH. A vertex cover must contain one end of every edge in a matching. Thus for all graphs $\beta(G) \geq \alpha'(G)$. To prove equality for a bipartite graph with bipartite sets X and Y , let Q be a minimum vertex cover. Partition the graph into four sets $Q_X = Q \cap X$, $Q_Y = Q \cap Y$, $T_X = X - Q_X$ and $T_Y = Y - Q_Y$.

We claim that in the bipartite graph induced by $Q_X \cup T_Y$ there is a matching that saturates Q_X . For consider $S \subseteq Q_X$ and $N(S) \cap T_Y$. The key observation is that the union of $Q - S$ and $N(S) \cap T_Y$ is a vertex cover of G : all edges that aren't covered by $Q - S$ have an end in S and hence also have an end in $N(S) - Q$. Thus, if $|N(S) \cap T_Y| < |S|$ we have a contradiction of the minimality of Q . Hence we can apply Hall's theorem. Similarly in the graph induced by $Q_Y \cup T_X$ there is a matching that saturates Q_Y . So we have a matching of size $|Q_X| + |Q_Y|$, which is what we were trying to prove. $\square$

It is immediate that for if G is a graph of order n , then $\alpha(G) + \beta(G) = n$. Gallai showed that $\alpha'(G) + \beta'(G) = n$ as well.

Chapter 4: Connectivity

4.1 Cycles in 2-connected Graphs

We start with some useful facts about 2-connected graphs.

WHITNEY. *A graph with at least three vertices is 2-connected if and only if every pair of vertices are in a cycle.*

PROOF. (1) Assume every pair of vertices lie in a common cycle. Let v be a vertex and consider any two other vertices x and y . The condition implies there is a cycle containing x and y , and hence there are two paths between x and y with no internal vertex in common. The removal of a vertex v can affect only one of the paths. Thus v is not a cut-vertex. Since this is true for every vertex v , there does not exist a cut-vertex. That is, the graph is 2-connected.

(2) Assume the graph is 2-connected. Let x and y be distinct vertices. We claim that there is a cycle containing x . For, we observed earlier that being 2-connected implies every vertex has degree at least 2. If u_1 and u_2 are neighbors of x , then in $G - x$ there is still a path between u_1 and u_2 ; this combined with the u_1, x, u_2 path produces a cycle containing x .

Now, out of all cycles containing x , choose the cycle C such that the distance from y to the cycle is as small as possible. Suppose, for sake of contradiction, that y is not on C . Consider a shortest path from C to y ; say it starts with vertices a and b (where possibly $b = y$). By the 2-connectendess assumption, there is a path from b to $C - a$ in the graph $G - a$. This path can be combined with part of the cycle C to produce a cycle C' that contains both x and b . But this cycle contradicts the original choice of C , since y is at least one closer to C' . Thus it must be the case that C contains y , as required. $\square$

Observe that if a graph is 2-connected then subdividing any edge yields a graph that is still 2-connected.

CYCLES THROUGH PAIRS. *Assume graph G has no isolates. There is cycle through each pair of vertices in G if and only if there is cycle through each pair of edges in G .*

PROOF. (1) Assume G has a cycle through each pair of edges. Then to get cycle through specified pair of vertices just choose an edge incident with each vertex.

(2) Assume G has a cycle through each pair of vertices. By previous result this means G is 2-connected. For each of the specified pair of edges e and f , subdivide the edge with vertex v_e and v_f . The result is still 2-connected and so one can find a cycle containing v_e and v_f . This cycle necessarily uses the two edges incident with v_e and with v_f ; so if we erase v_e and v_f we have a cycle using edges e and f . $\square$

4.2 Disjoint Paths

We say a collection of paths are *internally disjoint* if their internal vertices do not appear on any other path. For example, two vertices are joined by two internally disjoint paths if and only if they are in a common cycle.

MENGER. *If x and y are nonadjacent vertices, then the minimum size of an x - y cut is equal to the maximum number of internally disjoint x - y paths.*

PROOF SKETCH. If the graph consists of only x , y , and their neighborhoods, then use König-Egerváry to prove there is a matching between $N(x)$ and $N(y)$. Otherwise set things up to use induction. $\square$

Building on this, we have the following whose proof we omit:

WHITNEY. *A graph is k -connected if and only if there are k internally disjoint paths between every pair of vertices.*

A useful version (exercise):

FAN. *If a graph is k -connected, and $x, y_1, \dots, y_k$ are disjoint vertices, then there are k internally disjoint paths P_i where P_i goes from x to y_i .*

From this we can prove:

DIRAC. *For $k \geq 3$, if a graph is k -connected then every k vertices lie in a common cycle.*

PROOF. Fix some subset B of k vertices. By the earlier result, we know there is a cycle containing at least two of B . Let C be a cycle that contains the maximum number of B . Suppose this is not all of B ; that is, there is a vertex x of B not on C . Then, by the Fan theorem (or rather an extension thereof), there are $p = \min\{|C|, k\}$ internally disjoint paths from x to C , ending at different vertices of C . The p endpoints divide C into p segments. And by counting, there must be a segment that does not contain a vertex of B in its interior. The vertex x can be spliced into C replacing the segment, without losing any existing vertices of B , and thus we get a contradiction of the maximality of C . $\square$

There are similar results for edge-connectivity.

4.3 More

Graphs with bridges cannot have a strongly connected orientation. There is a converse whose proof we omit.

ROBBINS. *A graph has a strongly connected orientation if and only if it is 2-edge-connected.*

A network flow problem has a source, a sink and edge capacities, and the question is to maximize the flow from the source to the sink. Ford–Fulkerson: Max flow equals minimum cut. (Proof omitted. But sort of a generalization of Menger.) There is a connection with linear programming. Application: Network Flow can be used to compute the vertex and edge connectivity, and matching number.

Chapter 5: Vertex Colorings

5.1 Colorings

A coloring of a graph is a labeling of the vertices with labels/colors. A coloring is **proper** if every pair of adjacent vertices receive different colors; equivalently, the set of vertices of any particular color form an independent set. A graph is **k -colorable** if there is a proper coloring using a universe of size k . The **chromatic number** $\chi(G)$ is the least k such that the graph is k -colorable. For example, a graph being bipartite is equivalent to being 2-colorable; the Petersen graph is 3-colorable.

There is an immediate upper bound:

GREEDY. *For a graph of maximum degree Δ , the chromatic number is at most $\Delta + 1$.*

PROOF. Armed with a palette of $\Delta + 1$ colors, simply greedily color the graph. When come to color a vertex, one needs to give it a color that is different from each of the colors already present in its neighborhood. So at most Δ colors cause a problem, and there is at least one color available. $\square$

For a lower bound:

LOWER BOUND. (a) *The chromatic number is at least the clique number.*
(b) *The chromatic number is at least $n/\alpha(G)$, where n is the order and $\alpha(G)$ the independence number.*

PROOF. (a) Every vertex of the clique needs a different color.
(b) Every color can be used at most $\alpha(G)$ times. $\square$

A harder result is that the upper bound can be improved for most graphs; proof is given in next section.

BROOKS' THEOREM. *Except for cliques and odd cycles, $\chi \leq \Delta$.*

One useful fact, is that if have a coloring with the minimum number of colors, then for each color there must exist a vertex of that color that has neighbors of all other colors. (Otherwise we can re-color all vertices of that color.)

The *cartesian product* of two graphs G and H , written $G \square H$, is the graph with vertex set all pairs (u, v) of $V(G) \times V(H)$ where two pairs are adjacent if they are the same in one coordinate and adjacent in the other. For example: hypercube $Q_k = Q_{k-1} \square K_2$. The cartesian product has a spanning subgraph consisting of copies of G and one consisting of copies of H . These copies are called *fibers*.

CARTESIAN PRODUCT. $\chi(G \square H) = \max(\chi(G), \chi(H))$.

PROOF. Since the cartesian product contains both initial graphs, its chromatic number is at least the maximum of the two. On the other hand, say the maximum of the two is k and consider an optimal coloring g of G and h of H ; then assign (u, v) the color $g(u) + h(v) \bmod k$. One can then check that adjacent vertices in the product get different colors. $\square$

5.2 Proof of Brooks

We first prove the result for a 2-connected graph. The proof idea: do a greedy coloring but ensure when get to last vertex it has two neighbors of the same color. We need a “nice” spanning tree:

LEMMA. *If G is a 2-connected graph that is not complete nor a cycle, then there exists a pair of (nonadjacent) vertices x and y at distance two such that $G - \{x, y\}$ is connected.*

PROOF. Suppose there is a vertex u of degree $n-1$, where n is the order. Then since G is not complete, there exist two vertices x and y that are not adjacent, and it can be checked that the conditions are satisfied.

So assume no vertex has degree $n-1$. Since the graph is not a cycle, there is a vertex w that has degree more than 2. If $G - w$ is 2-connected, then let $x = w$ and y be a vertex at distance two from w , and it can be checked that the conditions are satisfied.

So assume $G - w$ has a cut-vertex. Just as a tree has two leaves, it can be shown that a graph with a cut-vertex has two end-blocks, meaning a 2-connected

subgraph or a K_2 that contains exactly one vertex that is a cut-vertex in the original graph. So say $G - w$ has end-blocks B_1 and B_2 . Since G was 2-connected, vertex w must have a neighbor in each block, say x in B_1 and y in B_2 . Then it can be checked that x and y are at distance two in G , and that $G - \{x, y\}$ is connected (noting that w has a neighbor other than x and y). $\square$

LEMMA. *If G is a 2-connected graph that is not complete nor an odd cycle then $\chi \leq \Delta$.*

PROOF. If G is an even cycle, then $\chi = \Delta = 2$. Otherwise, by the above lemma there is a pair of nonadjacent vertices x and y with a common neighbor w such that $G - \{x, y\}$ is connected.

Order the vertices of $G - \{x, y\}$ in decreasing distance from w as measured in $G - \{x, y\}$. Say the vertices are $v_1, v_2, \dots, v_{n-2}$ where $v_{n-2} = w$. Note that every vertex has a neighbor that is closer to w . So for each $i < n - 2$ there exists a $j_i > i$ such that v_i and v_{j_i} are adjacent.

Now we color G using the colors $\{1, 2, \dots, \Delta\}$. Start by coloring x and y with color 1. (They are not adjacent.) Then color the v_i in increasing order. For each v_i except the final one, when we color it at least one of its neighbors (namely v_{j_i}) is uncolored; thus there are at most $\Delta - 1$ colors among its neighbors, and so there is at least one color that we can assign to v_i . For w , all of its neighbors are colored before it; but two of them (namely x and y) get the same color and so there is at least one color that is not present among w 's neighbors that we can use for w . $\square$

Finally we prove the theorem for general graphs. The proof is by induction.

If G is 2-connected then we are done. So suppose the removal of vertex v disconnects G into two connected components. (Or more than two, it doesn't really matter.) Let G_1 and G_2 be the two connected graphs each consisting of one of these components together with the vertex v . Note that the chromatic number of G equals $\max\{\chi(G_1), \chi(G_2)\}$, since one can take an optimal coloring of each G_i and rename colors if needed so that v has the same color in both.

We claim that $\chi(G_i) \leq \Delta(G)$. If G_1 is neither an odd cycle nor complete, then this holds by the induction hypothesis. If G_1 is an odd cycle or complete,

then $\chi(G_1) = \Delta(G_1) + 1$. But $\Delta(G_1) < \Delta(G)$, since the vertex v has maximum degree in G_1 and its degree in G is bigger than its degree in G_1 .

5.3 Extremal Ideas

Mycielski construction: Can obtain triangle-free graphs with arbitrarily large chromatic number. The idea is to start with $G_2 = K_2$ and construct G_{i+1} from G_i by cloning all the vertices but having no edges between the new vertices, and adding a final vertex. The graph G_4 is known as the Grötzsch graph.

A graph is ***k-critical*** if the chromatic number is k but every proper subgraph can be colored with $k - 1$ colors. So the 3-critical graphs are the odd cycles. It is easy to show that the minimum degree of a k -critical graph is at least $k - 1$, and that it is 2-connected (for $k \geq 3$).

The above construction shows that having chromatic number large does not require a large clique. But we can say something:

DIRAC. *Every graph with chromatic number at least 4 contains a subdivision of K_4 .*

We saw Manvel's theorem earlier. Here is an extension that can be proved.

TURÁN. *The maximum number of edges in a graph that has no K_{k+1} is achieved by taking a complete k -partite graph with as equal parts as possible.*

Where a complete multipartite graph is the simple graph where the vertex set can be partitioned into subsets such that each subset is an independent set and any two vertices in different subsets are adjacent.

5.4 Chordal Graphs and Perfect Graphs

There are several ways to produce a graph by taking a collection of objects and making each object a vertex and defining adjacency if the two objects are related some how. For example, an ***interval graph*** is one that can be obtained from a collection of intervals that are adjacent if they overlap.

INTERVAL COLORING. *The chromatic number of the interval graph is its clique number.*

PROOF. Order the intervals by their left-most endpoint and run the greedy algorithm coloring the intervals. If one needs color k for a particular interval, then it overlaps an interval with each of color 1 through $k - 1$; but it and these intervals must form a clique. $\square$

An interval graph is a special case of chordal graphs. A **chordal graph** is defined to be a simple graph that does not have a chordless cycle (meaning an induced cycle of length at least 4). It can be shown that:

ELIMINATION ORDERING. *A graph is chordal if and only if there is an ordering of the vertices $v_1, v_2, \dots, v_n$ such that if we successively delete vertices in order, at the moment of deletion the neighborhood of v_i is complete.*

As a consequence we get that chordal graphs have chromatic number equal to clique number. (Do the greedy algorithm but process the vertices $v_n, \dots, v_1$.)

It can also be shown that:

CHARACTERIZATION. *A graph is chordal if and only if it is the intersection graph of subtrees of a tree.*

If a graph and all its induced subgraphs have chromatic number equal to clique number, then the graph is called **perfect**. A chordal graph is a special case of a perfect graph.

Chapter 6: Line graphs and Edge-coloring

6.1 Line Graphs

The **line graph** $L(G)$ of G has as vertex set the edges of G , with two vertices of $L(G)$ adjacent if the corresponding edges of G share an end-vertex. For example, a matching in G corresponds to an independent set in $L(G)$. It is easy to show that a line graph is **claw-free**, meaning it does not have an induced copy of $K_{1,3}$.

For each vertex v of the original graph G , the edges of G incident with v become a clique in $L(G)$. Further, each edge in G has two ends, and so is part of two cliques in $L(G)$. It turns out that the converse is true:

KRAUSZ DECOMPOSITION. *A graph H is the line graph of some simple graph if and only if one can decompose the edges of H into cliques such that every vertex is in at most two cliques.*

There are graphs that are claw-free but not line graphs, for example the wheel W_5 . In fact, Beineke showed that there are eight graphs that are minimal with respect to the property of being claw-free but not line graphs. This implies a polynomial-time algorithm for determining whether a graph is a line graph. (More on algorithms later.)

6.2 Edge-Chromatic Number

The edge-chromatic number (or chromatic index) χ' of a graph is the minimum number of colors needed to color the edges such that no two edges that share a vertex have the same color. This equals $\chi(L(G))$. It is immediate that $\chi'(G) \geq \Delta(G)$.

THE COMPLETE GRAPH. $\chi'(K_n)$ is $n - 1$ if n is even and n otherwise.

PROOF. Clearly $n - 1$ is a lower bound. It can be observed that an r -regular graph has $\chi' = r$ if and only if the edges can be decomposed into r perfect matchings. Therefore $n - 1$ is not possible when n is odd. The upper bound is left as an exercise. $\square$

König showed that if a graph is bipartite then $\chi'(G) = \Delta(G)$. (We already observed this for regular bipartite graphs.) A deeper result is:

VIZING/GUPTA. *For any simple graph G it holds that $\chi'(G) \leq \Delta + 1$.*

6.3 The List Chromatic Number

Suppose each vertex v of a graph has a list/palette L_v of colors. A valid coloring of a graph means give each vertex a color from its list such that the result is a proper coloring. The **list chromatic number** is then the smallest integer k such that for every possible collection of lists of size k , it is possible to find a valid coloring. Denoted χ_ℓ . The ordinary chromatic number corresponds to the case where the lists are all the same. Thus $\chi_\ell(G) \geq \chi(G)$.

For example, $\chi_\ell(C_5) = 3$. We say a graph is **k -choosable** if the list chromatic number is at most k . For example, the greedy algorithm shows that G is $(\Delta + 1)$ -choosable.

COMPLETE BIPARTITE. For example, $K_{3,3}$ is not 2-choosable. For each side, make one vertex have list $\{1, 2\}$, one list $\{1, 3\}$ and one $\{2, 3\}$. Then to color the vertices on one side one has to use at least two colors, which means four colors in total (since the colors on the two sides have to be disjoint), a contradiction. Also $K_{2,4}$ is not 2-choosable. It can be shown that $K_{7,7}$ is not 3-choosable: for each side, give each vertex one of the blocks/lines of the Fano plane (the decomposition of K_7 into K_3 's). The exact value of $\chi(K_{n,n})$ is in general not known, but is asymptotically $\log_2 n$.

The most major open problem is the edge-coloring version:

CONJECTURE. *For every graph, $\chi'_\ell(G) = \chi'(G)$.*

Galvin proved equality when the graph is bipartite.

Chapter 7: Planar Graphs

7.1 Plane Graphs

A **plane graph** is a graph embedded in the plane such that no pair of lines intersect. The graph divides the plane up into a number of regions called **faces**. A graph is **planar** if it has a planar embedding. Here are embeddings of K_4 and $K_{2,3}$.



For example, a plane graph has one face iff it is a forest. Further, if a graph is 2-connected, then every face is bounded by a cycle. This uses Jordán's curve theorem, which is actually more difficult to prove than it looks.

EULER'S FORMULA. $V - E + F = 2$ for connected plane graph with V vertices, E edges and F faces.

PROOF. By induction on E . If $E = V - 1$, then G is a tree and we're done. Otherwise $E \geq V$ and there is a cycle containing some edge e . The removal of e merges two faces. $\square$

It follows that:

BOUND. For plane graph, $E \leq 3V - 6$

PROOF. Let M be the number of edge-face pairs. Each edge appears twice. Each face appears at least three times. So we get $2E = M \geq 3F$. That is, $F \leq 2E/3$. Plug, into formula and algebra. $\square$

As a consequence, we get two important facts: (i) that K_5 is not planar, and (ii) that a planar graph has minimum degree at most 5. A **maximal planar graph** is one where every face is a triangle; equivalently one where $E = 3V - 6$.

If a graph is bipartite, one can improve the bound, since every region has a boundary of size at least 4:

BIPARTITE PLANAR. *If G is bipartite, then $E \leq 2V - 4$.*

Consequence: $K_{3,3}$ is not planar.

7.2 Duals, Drawings, and Outerplanar Graphs

The **dual** of a plane graph is obtained by placing a vertex inside each face, and joining two such vertices if the faces are adjacent. Note that the dual has the same number of edges as the original. Indeed, each edge e has a partner e^* in the dual. If the graph is suitably connected, then the dual of the dual is the original. It can be shown that a set of edges forms a cycle in the graph if and only if the set of dual edges form a minimal cut in the dual.

If we wrap a drawing onto a sphere, and then off again, we can move any face to be the exterior face. Every planar graph has an embedding such that all edges are line segments. We omit the proofs of the following:

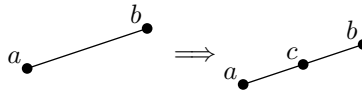
DRAWINGS. *If G is 3-connected, then:*

- (a) the face boundaries are its non-separating induced cycles*
- (b) any two planar embeddings are “equivalent”.*

A graph is **outerplanar** if it has an embedding in the plane with all the vertices on the outer face. It can be shown that neither K_4 nor $K_{2,3}$ is outerplanar. Also that the minimum degree of outerplanar graph is at most 2. And that its dual is the join of K_1 and a tree.

7.3 Kuratowski and Wagner

Recall that a **subdivision** of a graph is obtained by adding vertices of degree two into edges. Clearly, taking a subdivision of a graph preserves whether it is planar or not.



KURATOWSKI'S THEOREM. *A graph is planar if and only if it does not contain a subdivision of either K_5 or $K_{3,3}$.*

PROOF SKETCH. We saw earlier that K_5 and $K_{3,3}$ are not planar. So neither is any of their subdivisions.

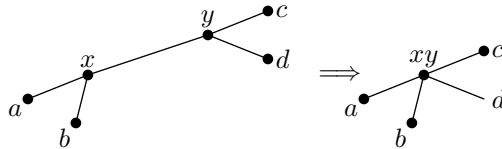
Suppose G is not planar. We may assume G is minimal nonplanar; that is, removal of any edge or vertex yields a planar graph. In particular it follows that G is 2-connected. (In general a graph is planar if and only if all its blocks are planar.) Further, we may assume that G is not the subdivision of another graph.

Now, we claim G has minimum degree at least 3. Suppose vertex v has degree 2 with neighbors x and y . If x and y not already adjacent, then splice v out—delete v and add edge xy . Then resultant graph is such that G is subdivision thereof, a contradiction of the minimality of G . And if x and y are already adjacent, then one can delete v , embed $G - v$ and add back in v in either region where xy is a boundary. Thus the claim is proven.

Now, it follows that there is an edge $e = uv$ such that $G - e$ is 2-connected. (Proof omitted.) By the assumption on G , the graph $G - e$ is planar.

Consider a cycle C of $G - e$ containing u and v . We would like to add edge e . So we may assume that there are barriers to doing this inside and outside C . If we make the interior of C as large as possible, we can assume properties outside C . Then argue that no matter how place inside C , one obtains a K_5 or $K_{3,3}$ subdivision. (Details omitted.) $\square$

The **contraction** of edge e is to delete the ends of e and build a new vertex adjacent to all vertices originally adjacent to one or both the ends of e . A graph G is a **minor** of H if G can be obtained from H by deleting vertices and contracting edges. It is straightforward that: If G is a subdivision of H , then H is a minor of G . Again, contractions preserve planarity; that is, if G is planar then so is any contraction of G .



WAGNER'S THEOREM. *A graph is planar if and only if it contains either K_5 or $K_{3,3}$ as a minor.*

One direction is a consequence of Kuratowski's theorem: if G is not planar it contains a subdivision of one of the graphs and hence a minor. On the other hand, if G is planar, then all its minors are planar; so it cannot contain either K_5 or $K_{3,3}$ as a minor.

7.4 Chromatic Number

We noted that by Euler's formula, every planar graph has vertex of degree at most 5. It follows that every planar graph is 6-colorable.

Every planar graph is 4-colorable. Kempe thought he proved this. But proof wrong; his approach was salvaged to prove that every planar graph is 5-colorable:

KEMPE CHAINS. *Every planar graph is 5-colorable.*

PROOF. The proof is by induction on the order. We noted above that every planar graph G has a vertex v of degree at most 5. The graph $G - v$ is 5-colorable by the induction hypothesis. One can re-insert vertex v , and color it, unless it has a neighbor of each color.

Consider a planar drawing of G where vertex v has neighbors v_1 through v_5 in cyclic order; assume each v_i has color i . Let $H_{i,j}$ be the subgraph of $G - v$ induced by the vertices of colors i or j . If vertices v_1 and v_3 are not connected in $H_{1,3}$, then one can interchange colors 1 and 3 in the component thereof containing v_1 , and thus get to situation where v has no neighbor of color 1; whence we are done. So we may assume that vertices v_1 and v_3 are connected in $H_{1,3}$. Similarly, we may assume that vertices v_2 and v_4 are connected in $H_{2,4}$. But these two connections contradict the planarity. $\square$

The famous 4-color theorem of Appel, Haken, and Koch used computers and showed that every planar graph is 4-colorable. (Robertson, Sanders, Seymour, Thomas redid.)

The elegant result is by Thomassen:

PLANAR CHOOSABLE. *Every planar graph is 5-choosable.*

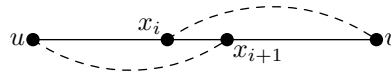
Chapter 8: Hamilton Cycles and Beyond

We define a **Hamilton cycle** as one that contains all the vertices; similarly a **Hamilton path** is one that contains all the vertices.

DIRAC'S THEOREM. *If G is a graph with order $n \geq 3$ and minimum degree $\delta \geq n/2$, then G contains a Hamilton cycle.*

PROOF. Suppose there is a counterexample, meaning a graph with the minimum degree condition that does not have a hamilton cycle. Then consider the counterexample G with the maximum number of edges. Clearly G is not complete; say it is missing edge uv . So consider the graph $G + uv$. By our discussion, this has a Hamilton cycle, and that Hamilton cycle uses edge uv . So in G there is a Hamilton path P from u to v .

Say P is $u = x_1, \dots, v = x_n$. Suppose there is i with $2 \leq i \leq n - 2$ where the pair $Q_i = \{ux_{i+1}, vx_i\}$ of edges both exist; then we get a cycle using all vertices of G .



The union of the Q_i accounts for all the edges incident with u or v , except for ux_2 and vx_{n-1} . Thus the total number of edges incident with them is at most $n - 1$. But that contradicts the minimum degree condition. $\square$

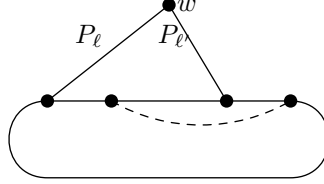
This theorem is best possible. Here are two graphs with minimum degree $(n - 1)/2$ that do not have a Hamilton cycle: $K_{m,m+1}$; or two copies of K_m with one vertex of each identified. There are many extensions. The first was by Ore: if the degree-sum $d(u) + d(v) \geq n$ for all nonadjacent vertices u and v , then the graph has a Hamilton cycle.

CHVÁTAL-ERDŐS. *If G is a graph with order at least 3 and the connectivity is at least the independence number, then the graph has a Hamilton cycle.*

PROOF. If the independence number is 1, then the graph is complete. So assume the independence number $k \geq 2$. In particular this means the connectivity is at least 2. Consider the longest cycle C . Suppose there exists a vertex w outside the cycle. By Menger's Theorem, there are internally disjoint paths $P_1, \dots, P_k$

from w to C . If any two paths meet C at consecutive vertices of C , then one can insert w to get a longer cycle. So assume not.

For each path P_ℓ , let v_ℓ be the vertex of C after the end of P_ℓ going clockwise. We claim that: the set $\{v_\ell\}$ is an independent set. This follows because a chord $v_\ell v_{\ell'}$ would enable a longer cycle using the chord, P_ℓ , and $P_{\ell'}$.



But by the condition of theorem, adding w to $\{v_\ell\}$ yields a set that is not independent. This means that w is adjacent to some v_ℓ , and one can thus insert w using P_ℓ and this edge. This contradicts the claim that C is a longest cycle.

□

For planar graphs, Tutte showed that 4-connected implies the existence of a Hamilton cycle. In general, a necessary condition for a Hamilton cycle is that the removal of any vertex set S leaves at most $|S|$ components. The **toughness** of a graph is defined to be the minimum of the ratio $|S|/k(G - S)$ over all cut-sets S , where $k(G - S)$ is the number of components of $G - S$.

OPEN QUESTION. *Does a sufficiently large toughness guarantee a Hamilton cycle?*

A graph is **pancyclic** if it contains cycles of each length from 3 up to its order. The minimum-degree threshold to guarantee a graph is pancyclic is only slightly larger than Dirac's bound:

MINIMUM DEGREE. *If G is a graph with order $n \geq 3$ and minimum degree $\delta \geq (n + 1)/2$, then G is pancyclic.*

However, if the graph is known to have a Hamilton cycle, then the threshold can be reduced:

BFG. *If G is a graph that contains a Hamilton cycle and a triangle and minimum degree $\delta \geq (n + 2)/3$, then G is pancyclic.*

Part II

Core

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Chapter 9: Products and Cayley Graphs

9.1 Product Graphs

There are multiple ways to combine two graphs to get a new one. For a product of graphs G and H , the vertex set is the set of ordered pairs (g, h) where $g \in V(G)$ and $h \in V(H)$. The most common product is the **cartesian product** written $G \square H$. Here two pairs are adjacent if they are the same in one coordinate and adjacent in the other. For example: hypercube $Q_k = Q_{k-1} \square K_2$. The cartesian product has a spanning subgraph consisting of copies of G and one consisting of copies of H ; these copies are called its **fibers**. The most famous result about cartesian products is the formula for their chromatic number: $\chi(G \square H) = \max(\chi(G), \chi(H))$.

The product $G \times H$ has several names including **direct product** and **tensor product**. (Note that some people use notation $\times$ for cartesian product.) In the direct product two pairs are adjacent if both their first and second coordinates are adjacent. Note for example that if both G and H are connected nontrivial bipartite graphs, then $G \times H$ is bipartite but has two components. And in general $\chi(G \times H) \leq \min(\chi(G), \chi(H))$. For 50 years it was conjectured that $\chi(G \times H) = \min(\chi(G), \chi(H))$, but this was disproved by Shitov.

Another product $G \boxtimes H$ also has several names including **strong produce** and **compositioin**. Here two pairs are adjacent if their first coordinates are adjacent, or if they are the same in the first coordinate and adjacent in the second. Essentially one takes G and replaces each vertex by a copy of H ; the resultant edges are the union of the cartesian and direct products.

9.2 Cayley Graphs

Let Γ be a group and S a subset of Γ . Then the **Cayley graph** generated by S has vertex set Γ ; and an arc (α, β) is present if and only if $\beta = \alpha\sigma$ for some $\sigma \in S$. That is, $\alpha^{-1}\beta \in S$. For example, if we take $\mathbb{Z}_n$ with set $S = \{1\}$, then the resultant Cayley graph is just the (directed) n -cycle. We assume the identity is not in S , so there are no loops. If S generates Γ , then the Cayley graph is

strongly connected. If S is closed under inverses, then one can consider the result an undirected graph. Note that every Cayley graph is regular.

The hypercube is an example of a Cayley graph. Indeed, if G and H are Cayley graphs, then their Cartesian product is also a Cayley graph, namely of the direct product of the groups for G and H .

Cayley graphs have an immediate symmetry. A graph is called **vertex-transitive** if all vertices are identical. That is, for every pair of vertices u and v there is an automorphism of the graph that maps u to v .

THEOREM. *A Cayley graph is vertex-transitive. Indeed, for any $\lambda \in \Gamma$, the mapping $\alpha \rightarrow \lambda\alpha$ is an automorphism.*

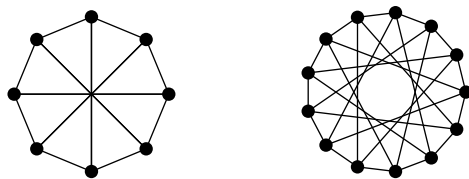
PROOF. We know (α, β) is an arc if and only if $\alpha^{-1}\beta \in S$. Thus, $(\lambda\alpha, \lambda\beta)$ is an arc if and only if $(\lambda\alpha)^{-1}(\lambda\beta) \in S$. But $(\lambda\alpha)^{-1}(\lambda\beta) = \alpha^{-1}\beta$. $\square$

In fact, it follows that the automorphism group of the Cayley graph has a subgroup isomorphic to Γ . An undirected Cayley graph also has connectivity properties: for example if it is connected then its edge-connectivity equals its degree.

A nonexample is the Petersen graph. Though the Petersen graph is vertex-transitive, it is not a Cayley graph. The Petersen graph is a special case of the **Kneser graphs**, most of which are not Cayley. A Kneser graph is defined by taking all k -element subsets of an n -element universe, and making two subsets adjacent if and only if they are disjoint. The complement of a Kneser graph is a **Johnson graph**. An example of a Johnson graph is the line graph $L(K_m)$, which for even $m \geq 6$, is not Cayley.

9.3 Circulants and Paley Graphs

A **circulant graph** is where Γ is the cyclic group. Note that both K_n and $K_{m,m}$ are circulants. We can denote an undirected circulant by $C_n[a_1, \dots, a_r]$ where $S = \{\pm a_1, \dots, \pm a_r\}$. For example, here are the circulants $C_8[1, 4]$ (also called a Möbius ladder) and $C_{13}[1, 5]$.



A famous circulant is the ***Paley graph***. Let p be a prime power such that $p \equiv 1 \pmod{4}$. (Note that for such p it follows that -1 has a square-root.) Then define the circulant on p vertices with S the set of quadratic residues; that is, $S = \{a^2 \bmod p : 1 \leq a \leq p-1\}$. For example, the Paley graph on 5 vertices is the 5-cycle, while the Paley graph on 13 vertices is $C_{13}[1, 3, 4]$. One can show that the Paley graph is self-complementary (isomorphic to its complement). (If one takes a prime power such that $p \equiv 3 \pmod{4}$ then one gets a Paley digraph.)

Paley graphs are famous as examples of ***quasi-random graphs***: that is, explicit constructions that have “all” the properties of random graphs. In particular they are what are called ***expander*** graphs.

Chapter 10: Ramsey Numbers

10.1 Ordinary Ramsey Numbers

Define $R(m, n)$ as the minimum cardinality N such that every coloring of the edges of the complete graph K_N with red and blue yields either a red K_m or a blue K_n . A trivial example is that $R(2, m) = m$.

EXAMPLE. $R(3, 3) = 6$. Consider any vertex v in a coloring of K_6 . It has at least three edges of same color, say red, going to set S . Then either the set S is all blue, and we're done, or S has a red edge, which extends to a red clique with v . The lower bound is shown by having the red edges be C_5 .

EXAMPLE. $R(3, 4) = 9$. The lower bound is shown by the circulant $C_8[1, 4]$, which has clique number 2 and independence number 3. To show the upper bound, note that one cannot construct a 3-regular graph on 9 vertices; so there must be either a vertex u with 4 red neighbors or a vertex v with 6 blue neighbors. In the former, we get either a red edge or a blue K_4 in u 's red neighborhood. In the latter, we get a monochromatic triangle in v 's blue neighborhood.

The above example proves the idea for the general upper bound:

RECURRENCE. $R(m, n) \leq R(m - 1, n) + R(m, n - 1)$.

(Can be shown to be strict if both terms on the right-hand-side are even.)

Corollary: $R(m, n) \leq \binom{m+n-2}{m-1}$. We show later that $R(m, m) > 2^{m/2}$.

Some more small numbers: $R(3, 5) = 14$ and the extremal graph is the circulant $C_{13}[1, 5]$. $R(4, 4) = 18$ and the extremal graph is the Payley graph on 17 vertices.

10.2 Graph Ramsey Numbers

One can generalize from cliques to general graphs: For graphs G and H , define $R(G, H)$ as the minimum cardinality N such that every coloring of the edges of

K_N with red and blue yields either a red G or a blue H . Note that the copies of G and H do not have to be induced; thus $R(G, H) \leq R(m, n)$ where m and n are the orders of G and H .

Here are some elegant results.

THEOREM. *For any tree T on m vertices, it holds that*

$$R(T, K_n) = (m - 1)(n - 1) + 1.$$

PROOF. Lower bound: Color $K_{(m-1)(n-1)}$ so that the red edges form $(n - 1)$ disjoint copies of K_{m-1} .

Upper bound: induction. The base case $m = 2$ or $n = 2$ is easily checked. Now consider an end-vertex u in T with neighbor v . Let $T' = T - u$. By the induction hypothesis we can find either a red T' or a blue K_n . If the latter happens we are done; so assume the former. Let w be the vertex corresponding to v . If w has a red neighbor outside the copy of T' , then we are done. So we may assume w has blue-degree at least $(m - 1)(n - 2) + 1$. Now we induct in w 's blue neighborhood to get either a red T , or a blue K_{n-1} to which w is joined. $\square$

THEOREM. $R(mK_2, mK_2) = 3m - 1$.

PROOF. Lower bound: partition the vertex set of K_{3m-2} into a red clique of size $m - 1$ and a blue clique of size $2m - 1$ and color red all the edges between the cliques.

Upper bound: Use induction. The base case is $m = 1$ which is trivial. If the coloring of K_{3m-1} is monochromatic, then we are done. So we may assume there exists red and blue edges r and b overlapping in some vertex. Delete the three vertices of r and b , and apply the inductive hypothesis. $\square$

10.3 Bipartite Ramsey Numbers

One can further change the host graph. Define the bipartite Ramsey number $B(m, n)$ as the minimum cardinality N such that every coloring of the edges of $K_{N,N}$ with red and blue yields either a red $K_{m,m}$ or a blue $K_{n,n}$. (There are other versions of bipartite Ramsey numbers.)

EXAMPLE. $B(2, 2) = 5$. Lower bound: One can decompose $K_{4,4}$ into two 8-cycles.

Upper bound: If one colors $K_{5,5}$ there are at least 13 edges of the same color. But it is impossible to have 13 edges in $K_{5,5}$ without a 4-cycle, as we show next.

Suppose one has a subgraph of $K_{5,5}$ with 13 edges but no 4-cycle. Say $K_{5,5}$ has bipartition (X, Y) . Let A be the number of P_3 's with centers in X . Then $A \leq \binom{5}{2} = 10$, since there can be at most one P_3 for every pair of vertices in Y . On the other hand, $A \geq \sum_{i=1}^5 \binom{d_i}{2}$, where $d_1, \dots, d_5$ are the degrees in X . Since these degrees sum to the number of edges, and the binomial coefficient as a function has positive second derivative, the best one can do is degrees 3, 3, 3, 2, 2, but in any event $A \geq 11$, a contradiction.

The maximum number of edges in a bipartite graph without a specified subgraph is called a **Zarankiewicz** number. These turn out to be helpful in bipartite Ramsey theory. Note that the normal equivalents, Turán numbers, are irrelevant to ordinary Ramsey numbers, since the complement of extremal graphs have huge clique number.

The next theorem is about 3-coloring the edges of a complete bipartite graph to avoid a monochromatic 4-cycle. It was proved by Exoo (and later re-proved by GHO).

THEOREM. $B(2, 2, 2) = 11$.

PROOF. It can be shown that $\lceil 121/3 \rceil = 41$ edges guarantee a 4-cycle in $K_{11,11}$. However, there is a unique subgraph of $K_{10,10}$ that has 34 edges and no 4-cycle. And one can find a suitable coloring of K_{10} (for example using a computer). $\square$

Chapter 11: Domination and Hypergraphs

11.1 The Domination Number of a Graph

A set S of vertices **dominates** a graph if every vertex outside the set S is adjacent to at least one member of the set S . The **domination number** $\gamma(G)$ of graph G is the minimum size of a dominating set.

For example, for the complete graph K_n we have $\gamma(K_n) = 1$ and for the cycle C_n we have $\gamma(C_n) = \lceil n/3 \rceil$. On a chessboard, the domination problem is related to the minimum number of pieces to attack all squares.

In general we say that each vertex **dominates** itself and its neighbors. It follows that each vertex dominates at most $1 + \Delta$ vertices, where Δ denotes the maximum degree. Thus one obtains the fundamental lower bound:

THEOREM. For a graph G of order n , $\gamma(G) \geq n/(1 + \Delta)$.

Equality in the bound implies that no two vertices of S are joined by an edge, and every vertex outside S has a unique neighbor in S . This is sometimes called an “efficient/perfect dominating set”. For example, two antipodal vertices in the 3-cube Q_3 form such a set. Indeed, in the cube in general, an efficient/perfect dominating set is equivalent to a perfect 1-error-correcting code.

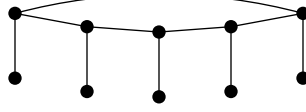
The fundamental upper bound for the domination number is:

THEOREM. If graph G has order n and no isolates, then $\gamma(G) \leq n/2$.

PROOF. A dominating set is **minimal** if removing any vertex from it implies that it no longer dominates. Consider a minimal dominating set S . We claim that every vertex of S has a neighbor outside S . Suppose some vertex $v \in S$ has no neighbor outside S . Since the graph has no isolates, v has a neighbor in S . But then when we remove v from S , every vertex is still dominated, a contradiction.

Thus every vertex of S has a neighbor outside S . Or, put another way, the complement of S is also a dominating set of G . This means that G has two disjoint dominating sets, and so one of them has at most half the vertices. $\square$

It can be proved that the connected graphs with equality in the above bound are (a) C_4 ; or (b) the corona of a graph. The **corona** of a graph is obtained by adding for each vertex v a new end-vertex v' adjacent only to v . For example, here is the corona of C_5 .



The interesting open problem here is Vizing's conjecture for the Cartesian product:

CONJECTURE. *For all graphs G and H it holds that $\gamma(G \square H) \geq \gamma(G) \gamma(H)$.*

Another open question is: find the best possible upper bound for the domination number of a cubic (3-regular) graph.

11.2 Nordhaus–Gaddum Results

Early on Nordhaus and Gaddum published a paper about relationships between the chromatic number of a graph and the chromatic number of its complement. In particular they showed that the sum of the chromatic numbers is at most $n+1$ where n is the order. Results of such kind have become known as Nordhaus–Gaddum results. Here are the ones for domination, proved by Jaeger and Payan:

THEOREM. *For $n \geq 2$:*
 (a) $3 \leq \gamma(G) + \gamma(\bar{G}) \leq n + 1$
 (b) $2 \leq \gamma(G) \times \gamma(\bar{G}) \leq n$.

PROOF. Lower bound: except when $n = 1$, it is impossible to have both G and $\bar{G}$ having $\gamma = 1$. Thus one obtains the two bounds.

Upper bound on sum: To get a dominating set of G , take vertex v and its non-neighbors. To get a dominating set of $\bar{G}$, take vertex v and its non-neighbors in $\bar{G}$. So we take v twice and every other vertex at most once. (Attained for complete graph.)

Upper bound on product: Let $S = \{v_1, \dots, v_\gamma\}$ be a minimum dominating set of G . Each vertex in the graph is dominated by at least one vertex in S .

Thus one can partition $V(G)$ into γ disjoint sets each containing one vertex of S and some of its neighbors. Say the sets in the partition are T_i where T_i contains v_i and is a subset of $N[v_i]$ (the **closed neighborhood** of v_i , meaning it and its neighbors). Call a vertex in T_i **good** if it dominates T_i . (In other words it can substitute for v_i .) Out of all such partitions, choose the partition such that the number of good vertices is maximized.

Now, consider any of the T_i . We claim that T_i is a dominating set of the complement $\bar{G}$. For suppose it is not. Then there exists vertex w that in $\bar{G}$ is not dominated by T_i . That is, in G , vertex w is adjacent to all of T_i . Say $w \in T_j$. If w is good, then it dominates T_j in G and so $T - \{v_i\}$ dominates G , a contradiction to S being minimum. If w is not good, then this means that moving w to T_i turns it into a good vertex, which contradicts the choice of partition above. In other words, every T_i is a dominating set of $\bar{G}$ and hence has size at least $\gamma(\bar{G})$. It follows that $\gamma \times \gamma(\bar{G}) \leq \sum_{i=1}^{\gamma} |T_i| = n$, and the bound follows.

It is known that there is equality in the product upper bound if and only if G or $\bar{G}$ is: (i) complete; (ii) the union of C_4 and coronas; or (iii) the cartesian product $K_3 \square K_3$. $\square$

11.3 Independent Domination

There are two prominent variations on domination. The first is to require that the dominating set S is an independent set. The independent domination number $i(G)$ is the minimum size of a dominating set of G that is also an independent set. Equivalently, the minimum size of a **maximal independent set**.

For example: $i(K_n) = 1$; and $i(K_{m,n}) = m$ if $m \leq n$. For the chessboard, we are attacking all squares with mutually nonattacking pieces.

It follows that $i \geq \gamma$. There is no reverse inequality. For example, one can build a graph with $\gamma = 2$ and i arbitrarily large. And there is no nice upper bound; one can build a graph with $i(G) = n - o(n)$.

One nice result is the following, due to Allan and Laskar. Recall that a **claw** is an induced copy of $K_{1,3}$.

THEOREM. If graph G is claw-free, then $\gamma(G) = i(G)$.

PROOF. Out of all minimum dominating sets S , take an S such that the number of edges inside S is as small as possible. We claim that that number is zero; that is, S is an independent set.

Suppose that there is an edge inside S , say xy . Then the fact that we need x in S means there is some vertex z outside S that is dominated only by x . (Sometimes called an *external private neighbor*.)

Let set T be obtained from S by replacing x by z . Note that T has less internal edges than S . The only vertices that T might not dominate are those vertices like z that were originally dominated only by x . But these vertices must form a clique, since otherwise x is the center of a claw with y as one end. Thus z dominates all of these vertices. That is, T is a dominating set of G of the same size as S but with fewer internal edges, a contradiction of the choice of S . That is, S is an independent dominating set. $\square$

It is conjectured that for order $n > 10$, a cubic graph has independent domination number at most $3n/8$.

11.4 Total Domination

The other prominent variation of domination is to require that every vertex in S has a neighbor in S . Such a set S is called a *total dominating set*, and the minimum size is denoted $\gamma_t(G)$.

For example: $\gamma_t(K_n) = 2$; and $\gamma_t(C_n) = n/2$ if n is a multiple of 4. Note that $\gamma \leq \gamma_t$. There is a partial converse in that $\gamma_t \leq 2\gamma$. (Add a mate for every vertex in the minimum dominating set.)

Here the fundamental upper bound is due to Cockayne, Dawes, and Hedetniemi:

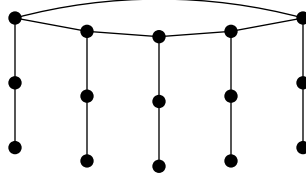
THEOREM. For a graph G of order $n \geq 3$ without isolates, $\gamma_t(G) \leq 2n/3$.

PROOF. We may assume G is a tree, because removing edges can only make one's job harder.

Further, if G has a bridge whose removal leaves two components each with at least 3 vertices, then one can induct. So assume G has no such bridge. In particular, this means that G has diameter at most 4.

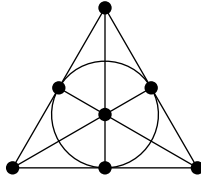
If G has diameter at most 2, then the central vertex dominates, and if diameter 3 then the central pair dominate. In either case $\gamma_t(G) \leq 2$. If G has diameter 4, then by results on trees we know that G has radius 2 and a single central vertex v . It follows that every non leaf-vertex has a leaf neighbor, except possibly v . Thus the set A of non-leaf vertices in G is a total dominating set, and has size at most $(n + 1)/2 \leq 2n/3$. $\square$

There is an infinite family of graphs with equality: take a corona and subdivide each leaf edge once. Here is a picture.



11.5 Transversals of Hypergraphs

A **hypergraph** has a set V of vertices, and a set E of hyperedges. Each hyperedge is a subset of V . In a graph, every hyperedge has size 2. A famous example of a hypergraph is the Fano plane. This is 3-regular (meaning every vertex is in 3 edges) and 3-uniform (meaning every hyperedge has 3 vertices).



A **transversal** of a hypergraph is a set of vertices that intersects every hyperedge. For a graph, a transversal is a vertex cover. Questions about total domination in graphs can be translated into questions about transversals of hypergraphs as follows. The open neighborhood of a vertex in a graph is the set of its neighbors. For a graph G , the **open-neighborhood hypergraph** $ONH(G)$ is the hypergraph with the same vertex set as G and such that every open neigh-

neighborhood in G is a hyperedge in $ONH(G)$. In particular, $ONH(G)$ has n vertices and n hyperedges. If G is r -regular, then $ONH(G)$ is r -uniform.

An interesting result about total domination, found by multiple authors, is that the total domination number of a cubic graph is at most $n/2$ (and there are examples of equality). One way to prove this is to show that the transversal number of a 3-uniform hypergraph with n vertices and n edges is at most $n/2$. This can be done by induction in a cute way, as shown by Thomasse and Yeo.

THEOREM. *A 3-uniform hypergraph has a transversal with size at most $(n + m)/4$ where n is the number of vertices and m the number of hyperedges.*

11.6 Connections

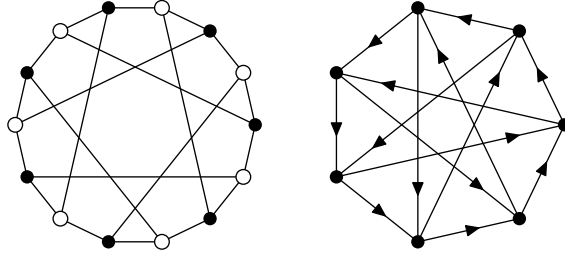
There is a deep connection among (a) set-systems/hypergraphs, (b) bipartite graphs, (c) directed graphs, and (d) matrices.

Given a hypergraph, one can construct the *incidence matrix* by numbering the hyperedges and vertices, and putting 0 or 1 depending on whether the vertex is in the hyperedge or not. One can then use the incidence matrix as the bipartite adjacency matrix of a bipartite graph: one partite set has a vertex for each row and one partite set has a vertex for each column. Note that this process is semi-reversible. If we start with the bipartite graph, then the open-neighborhood hypergraph has two components, one of which is the original hypergraph.

Furthermore, if the resultant bipartite graph has a perfect matching M with bipartition (X, Y) , one can form a digraph by the following process. For every edge not in M , orient it from X to Y . Then for every edge in M , contract it to a single vertex. Note that this process is reversible!

Consider for example the Fano plane. One gets the following matrix, and this yields the Heawood graph, and the Paley graph.

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$



11.7 Disjoint “Dominating” Sets

We saw every graph without isolates has two disjoint dominating sets. There is no minimum-degree condition that guarantees a graph has three disjoint dominating sets, nor two disjoint total dominating sets, nor two disjoint independent dominating sets. If one restricts to regular graphs, then there is such a minimum-degree condition for the first two cases.

One can again connect with hypergraphs. A graph having two disjoint total dominating sets means one can color the vertices red and blue such that every vertex has both a red and a blue neighbor. **Property B** for hypergraphs means coloring the vertices with two colors such that every hyperedge contains both colors. Thus the bipartite graph produced from a hypergraph has two disjoint total dominating sets if and only if the original set system has Property B.

It can be shown:

THEOREM. *Assume bipartite graph H is regular. Then H has two disjoint total dominating sets if and only if the associated digraph D has an even cycle.*

Thomassen proved that for every $k \geq 3$ every k -regular digraph has an even cycle; and thus for $k \geq 4$ every k -regular bipartite graph has two disjoint total dominating sets.

Chapter 12: Matrices and Graphs

There are many connections between matrices and graphs. We consider several here: the powers of the adjacency matrix, cages, counting perfect matchings, and properties of the eigenvalues of a graph.

12.1 The Adjacency Matrix and Counting Walks

The **adjacency matrix** A of a graph with n vertices is the $n \times n$ matrix with entry $a_{ij} = 1$ if vertex i and j are adjacent and 0 otherwise.

Recall that in a walk one can repeat vertices and edges. Number the vertices from 1 up to n , and let $w_k(i, j)$ be the number of walks from i to j of length k . For example, the number of walks from i to i of length 2 is the degree of i .

THE GOODWILL–HUNTING THEOREM. *The value $w_k(i, j)$ equals the (i, j) -entry of A^k , where A is the adjacency matrix.*

PROOF. We can prove this by induction on k . The base case is $k = 1$, which is true, since the adjacency matrix A^1 by definition gives the number of length-1 walks. In general, assume we know it is true for k and test for $k + 1$. Each walk from i to j of length $k + 1$ can be split into a walk of length k from i to some vertex ℓ followed by the edge ℓ to j . So the total number is

$$w_k(i, j) = \sum_{\ell=1}^n w_k(i, \ell)w_1(\ell, j) = \sum_{\ell=1}^n A_{i,\ell}^k A_{\ell,j} = (A^k A)_{i,j} = A_{i,j}^{k+1},$$

as required. $\square$

12.2 Cages

The next result is about the existence of particular graphs. Consider an r -regular graph with girth 5 (no triangle nor 4-cycle). Then a vertex, its neighbors, and their neighbors must all be distinct. This means there must be at least $1 + r + r(r - 1) = r^2 + 1$ vertices. The question is, when is there such a graph with exactly $r^2 + 1$ vertices? Such a graph is called a **cage**

For example, when $r = 2$ one has C_5 , and when $r = 3$ one has the Petersen graph. Using matrix algebra, we can almost solve this question.

THEOREM. *If there is an r -regular graph on $r^2 + 1$ vertices with girth 5, then r must be one of 2, 3, 7, or 57.*

PROOF. Let G be such a graph. Then the conditions mean that the diameter of G is 2. In fact, every two vertices are either adjacent or have a unique common neighbor. That is, if i and j are adjacent there is no walk of length 2 between them, and if i and j are not adjacent there is exactly one walk of length between them.

So by the above theorem this means that the matrix $A^2 + A$ has r 's on the diagonal and 1's everywhere else. As a matrix equation we have:

$$A^2 + A - (r - 1)I = J,$$

where J is the all-1 matrix.

Now let λ be an eigenvalue of A with eigenvector $\mathbf{v}$. The above equation implies that $\mathbf{v}$ is an eigenvector of J with eigenvalue $\lambda^2 + \lambda - (r - 1)$. A common exercise in a linear algebra class is to determine the eigenspaces of J . It can be checked that the matrix J has eigenvalue n with multiplicity 1 and the all-1-vector as eigenvector; and eigenvalue 0 with multiplicity $n - 1$.

The all-1 vector is an eigenvector of A with eigenvalue r . The above discussion means all other eigenvalues of A satisfy $\lambda^2 + \lambda - (r - 1) = 0$. This equation has roots

$$\lambda^+ = \frac{-1 + \sqrt{1 + 4(r - 1)}}{2} \quad \text{and} \quad \lambda^- = \frac{-1 - \sqrt{1 + 4(r - 1)}}{2}.$$

Now, suppose λ^+ has multiplicity b as eigenvalue of A . It follows that λ^- has multiplicity $n - 1 - b$. Recall from linear algebra that the sum of the eigenvalues with multiplicities is the trace, which for A is clearly 0. So we have that

$$b\lambda^+ + (r^2 - b)\lambda^- + r = 0.$$

Thus we have an equation for b , but b must be an integer. Some algebra later we get that r must be one of the values given in the statement of the theorem. $\square$

There is a graph known for the case $r = 7$. But the case $r = 57$ remains unresolved.

12.3 Perfect Matchings, Determinants and Permanents

Recall that the determinant of a matrix is given by a formula that involves the sum over permutations of the product of entries in the matrix:

$$\det A = \sum_{\pi} (-1)^{\text{sgn}(\pi)} \prod_i a_{i\pi(i)}$$

The **permanent** of A has the same formula except that the sign is deleted (all terms are added).

THEOREM. *For a graph G , the permanent of A has the same parity as the number of perfect matchings of G .*

PROOF. Each term $\prod_i a_{i\pi(i)}$ corresponds to a directed graph G_{π} where each vertex has in- and out-degree one and each arc is an orientation of some edge in G . Thus G_{π} is a collection of disjoint cycles, where some of the cycles have length 2. If G_{π} has a nontrivial cycle, then one can reverse the first such cycle to produce another graph. Thus we can pair off the π where G_{π} has a nontrivial cycle. We are left with the G_{π} that have only trivial cycles—but these correspond to a perfect matching. That is, the permanent of A has the same parity as the number of perfect matchings. $\square$

This led to Lovász's result:

THEOREM. *A graph has an even number of perfect matchings if and only if there is a non-empty set S of vertices such that every vertex in the graph (S included) has an even number of neighbors in S .*

PROOF. Consider a vector x and consider the product Ax . The i^{th} entry in the product is the sum of the values of x on the neighbors of the i^{th} vertex. It follows that if x is a 0-1-vector whose 1's correspond to S , that $Ax = 0$ in $\mathbb{Z}_2$ if and only if every vertex has an even number of neighbors in S . But by linear algebra, $Ax = 0$ has a nontrivial solution if and only if A has determinant zero. And working in $\mathbb{Z}_2$, the determinant and permanent are equal. And so the conclusion holds by the above theorem. $\square$

There are several other theorems that connect matchings to determinants.

12.4 Eigenvalues

The **adjacency matrix** A of a graph with n vertices is the $n \times n$ matrix with entry $a_{ij} = 1$ if vertex i and j are adjacent and 0 otherwise. The **eigenvalues** of the graph are the eigenvalues of the adjacency matrix. That is, $Ax = \lambda x$ for some nonzero vector x . By linear algebra, the eigenvalues are the roots of the characteristic polynomial, which is the determinant $\det(\lambda I - A)$. We deal only with the undirected graph, so the matrix is symmetric. Because the matrix is symmetric, the eigenvalues are real and there is a full set of eigenvectors.

If one views the vector x as weights on the vertices, then the i^{th} entry in Ax is the sum of the weights of the neighbors of the i^{th} vertex. So, for an eigenvector, one needs the sum of the weights of neighbors to be λ times the vertex's weight, for all vertices.

EXAMPLE. Consider the graph K_3 . The adjacency matrix has 0's on the diagonal and 1's everywhere else. Thus

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -1 & \lambda \end{vmatrix} = \lambda^3 - 3\lambda - 2.$$

The characteristic polynomial has roots 2 and -1 twice. If we put 1 on each vertex, we get eigenvector for $\lambda = 2$. If we use the three weights $\{0, 1, -1\}$ in any order, we have an eigenvector for $\lambda = -1$.

12.5 Regular Graphs and Bipartite Graphs

EXAMPLE. For the complete graph K_n the pattern for K_3 generalizes. If we put 1 on each vertex, we get an eigenvector for $\lambda = n - 1$. If we use weights $\{1, -1\}$ on two vertices and 0 elsewhere, we have an eigenvector for $\lambda = -1$. One can choose $n - 1$ linearly independent such vectors. (Why?) Thus the eigenvalues of K_n are $n - 1$ once and -1 $n - 1$ times.

The eigenvector generalizes for regular graphs:

FACT. *If graph G is r -regular, then the all-1 vector is eigenvector for $\lambda = r$.*

EXAMPLE. *The complete bipartite graph $K_{m,n}$. Let's try for an eigenvector where the weights are constant on each partite set; say the m vertices have weight a , and the n vertices have weight b . Then the requirement is that $\lambda a = nb$ and $\lambda b = ma$. Multiplying these together, we get $\lambda^2 ab = abmn$, so that $\lambda = \pm\sqrt{mn}$ is an eigenvalue. Okay: we must check that we can choose a and b nonzero, but that is easily checked. Also, 0 is an eigenvalue: put 1 and -1 on one vertex of the same partite set, and 0's elsewhere. One can do this $m + n - 2$ linearly independent ways (why?). Thus the eigenvalues of $K_{m,n}$ are $\sqrt{mn}$, $-\sqrt{mn}$, and 0 $m + n - 2$ times.*

This symmetry of the eigenvalues generalizes to all bipartite graphs:

FACT. *If graph G is bipartite, then the multiplicity of λ is the same as the multiplicity of $-\lambda$.*

The reason is that if ones takes an eigenvector for λ , and multiplies all weights in one partite set by -1 , one gets an eigenvector for $-\lambda$.

12.6 Further Results

EXAMPLE. *It is known that the eigenvalues of C_n are $2 \cos(2\pi r/n)$ for $0 \leq r < n$.*

THEOREM. *If graph G has eigenvalues $\{\lambda_i\}$ and graph H has eigenvalues $\{\mu_j\}$, then the cartesian product $G \square H$ has eigenvalues $\{\lambda_i + \mu_j\}$.*

PROOF. Let λ be an eigenvalue of G with eigenvector x , and μ an eigenvalue of H with eigenvector y . Construct a weighting of $G \square H$ by giving vertex (u_i, v_j) the weight $x_i \times y_j$. Then, the sum of the weights of the neighbors of vertex (u_i, v_j) is

$$\begin{aligned} \sum_{k \in N_G(u_i)} (x_k y_j) + \sum_{k \in N_H(v_j)} (x_i y_k) &= y_j \sum_{k \in N_G(u_i)} x_k + x_i \sum_{k \in N_H(v_j)} y_k \\ &= y_j \lambda x_i + x_i \mu y_j \\ &= x_i y_j (\lambda + \mu). \end{aligned}$$

Since x and y are both not the zero vector, this weighting is not all-zero. And so the result is an eigenvector for $\lambda + \mu$. Can then show that multiplicities carry over too. $\square$

For example, the eigenvalues of the hypercube Q_n are $n, n-2, n-4, \dots, -n$ with multiplicities given by binomial coefficients.

One can also obtain some information about the characteristic polynomial. Written as the form $\phi = \lambda^n + c_1\lambda^{n-1} + \dots + c_n$, it holds that $c_1 = 0$ and that c_2 is minus the number of edges. This comes from the determinant formula

$$\det H = \sum \left\{ (-1)^{\text{sgn}(\pi)} h_{i\pi(i)} : \pi \text{ is a permutation} \right\}.$$

Note that a permutation corresponds to a directed graph where we draw an arrow from each i to $\pi(i)$. Thus we get a nonzero contribution only when π corresponds to a union of a matching and directed cycles. This permutation formula then yields a recurrence for the characteristic polynomial:

$$\phi_G = \lambda\phi_{G-v} - \sum_{u \in N(v)} \phi_{G-v-u} - 2 \sum_{\text{cycle } C} \phi_{G-C}.$$

This follows by considering the permutations where $\pi(v) = v$, where $\pi(\pi(v)) = v$, and other cases. The factor of 2 is from the fact that a cycle has two orientations. This can be used, for example, to obtain a recurrence for the characteristic polynomial of the path and hence the cycle.

12.7 The Maximum and Second-Largest Eigenvalue

FACT. *If Δ is the maximum degree, then for all eigenvalues $-\Delta \leq \lambda \leq \Delta$.*

For, consider the vertex with the maximum absolute weight in an eigenvector. The absolute sum of its neighbors' weight is then at most its degree times its absolute weight. Indeed by results in linear algebra

$$\lambda_{\max} = \max\{x^t A x : x \text{ is unit vector}\}.$$

There is also an expansion property. For a set S of vertices, let $[S, \bar{S}]$ denote the edges with one end in S . The number of such edges is bounded by a function

of the second largest eigenvalue. We prove here only the result for a regular graph:

THEOREM. *If graph is k -regular with order n and second largest eigenvalue λ , then*

$$|[S, \bar{S}]| \geq (k - \lambda) |S| |\bar{S}| / n.$$

Note that if every edge leaving a vertex in S went to a random vertex, then the expected number of edges leaving S would be $k|S||\bar{S}|/n$.

PROOF. Let L be the Laplacian matrix $D - A$ where D is the diagonal matrix with the degrees of the vertices (here k). The Laplacian has smallest eigenvalue 0, and second smallest eigenvalue $\lambda - k$.

Now, consider $x^t L x$. Multiplying out we get that

$$x^t L x = x^t k I x - x^t A x = k \sum x_i^2 - 2 \sum_{i \sim j} x_i x_j = \sum_{i \sim j} (x_i - x_j)^2.$$

We will pick x such that it is orthogonal to the all-1 vector. Then, by a similar proof to the previous section, we have that

$$x^t L x \geq (k - \lambda) x^t x.$$

Indeed, set $x_i = s$ for $i \in S$ and $x_i = -(n - s)$ for $i \notin S$, where $s = |S|/n$. A few lines of calculation checks that such x is orthogonal to the all-1 vector, that $x^t x = s(n - s)n$, and that $\sum_{i \sim j} (x_i - x_j)^2 = |[S, \bar{S}]|n^2$. Putting this all together yields the result. $\square$

Chapter 13: NP-completeness

13.1 Problems in P

We consider “decision problems” for graphs. The input is a graph and maybe some other parameters (such as an integer k), collectively known as an “instance”, and there is a single yes/no question. Like: “Does the graph have an Eulerian circuit?” or “Does the graph have a clique of size k ?”

An *algorithm* is a set of clear instructions for solving a problem: the instructions break down into a sequence of steps. For an algorithm for a graph problem, let $f(n)$ denote the maximum number of steps that the algorithm would take when confronted with a graph on n vertices. The function $f(n)$ is known as the *running time* of the algorithm. Such an algorithm is said to be *polynomial-time* if there exists a polynomial $g(n)$ such that $f(n) \leq g(n)$ for all n . The set of all problems that can be solved in polynomial time is denoted by **P**. Colloquially people say such problems have “efficient” algorithms.

For example, consider the problem to determine if an input connected graph has an Eulerian circuit or not. Then a reasonable algorithm would simply calculate for each vertex the degree of that vertex and check that all degrees are even. Assume we are given the graph as an adjacency matrix. (The exact representation does not matter when we define P.) Then checking the parity of the degree of a particular vertex takes at most n steps. And the whole algorithm takes about n^2 steps. Thus there is an efficient algorithm to test whether a connected graph has an Eulerian circuit or not.

Common graph decision problems in P include:

- Is it bipartite?
- Is there a perfect matching?
- Is the distance between two specified vertices at most some specified value?
- Is the connectivity at least some specified value?

There are, of course, many graph problems where the answer is a value rather than just yes–no. We do not consider these explicitly, except to note that the

two types of problems often can be related. For example, if one can decide whether the connectivity is at least some threshold, then one can determine the actual connectivity by trying all thresholds (or to be more efficient using a binary search).

13.2 Problems in NP and NP-completeness

There are many graph problems for which efficient algorithms are not known. Suppose I know that the graph has a Hamilton cycle and I wish to convince you that there is one. Well I can simply tell you the Hamilton cycle and then you can easily verify that it is valid. But how would I prove to you that there was not a Hamilton cycle?

An instance of the problem for which the answer is yes is known as a yes-instance. For the problem of

INPUT: a graph. QUESTION: Does the graph have a Hamilton cycle?

there are short *certificates* or “proofs” for yes-instances.

A problem is in **NP** if there exists an efficient algorithm which, given an instance I and a potential certificate C , can tell whether the C gives a valid proof that the answer is yes. It is important to note that: If the instance is a yes-instance, then there must exist a certificate that will convince the algorithm. If the instance is a no-instance, then there must be no certificate that will convince the algorithm. Informally, we say that a problem is in NP if there exists an efficient algorithm which with some prompting can verify a yes-answer.

There are tens of thousands of graph problems in NP. For example,

- Does the graph have a Hamilton cycle?
- Is it 3-colorable?
- Is it decomposable into K_3 's?
- Is the independence number at least k ?

The certificate is, respectively, the cycle, the coloring, the decomposition, and the independent set.

Obviously $P \subseteq NP$. A major open question is whether $P = NP$ or not.

A problem is said to be **NP-complete** if the existence of an efficient algorithm for it would imply the existence of an efficient algorithm for every problem in NP. Essentially, the NP-complete problems are the hardest ones in NP. All four problems listed above are NP-complete. A problem is said to be **NP-hard** if it is at least as hard as all problems in NP.

13.3 Proving a Problem NP-Complete

Assume one has a new problem, say X , that is in NP. To prove that X is NP-complete, one provides a **reduction** from some known NP-complete problem, say K , to X . A reduction transforms a question in the language of K into the language of X without changing the answer. Several examples are provided below.

Note that one has complete freedom in the choice of K . A common choice is called SAT. For this problem the input is a boolean formula (meaning each variable can be either true or false) and the question is whether the formula is **satisfiable**: Is there a setting of the variables that makes the formula true. The formula is given in “conjunctive normal form”: this means it is the AND of a collection of **clauses**, where each clause is the OR of a collection of variables and negated variables, collectively called **literals**. An example formula is $(x \vee y \vee \bar{z}) \wedge (x \vee \bar{y} \vee w)$. This is easily satisfiable: for example when all variables are true, both clauses are true, and hence the formula is true.

13.3.1 Domination: Proof One

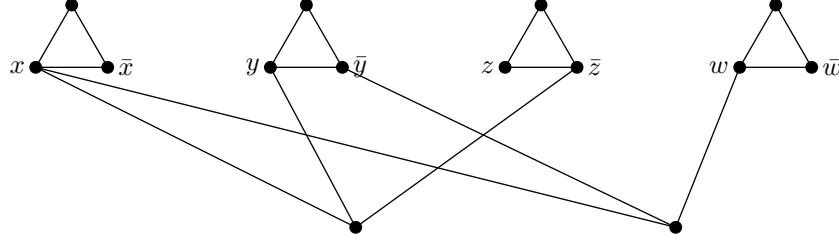
The DOMINATION problem has input a graph G and an integer k , and the answer is whether the graph has a dominating set of size at most k .

THEOREM. *The DOMINATION problem is NP-complete.*

The classical way to show this is to reduce SAT to it. When reducing SAT to a graph problem, it is common to create little subgraphs, called **gadgets**, for each variable, and for each clause, and then to join these up in some fashion.

Here is the reduction to DOMINATION: Consider the boolean formula ϕ . For each variable, say x , construct a triangle with two vertices labeled x and $\bar{x}$. For each clause, add a vertex. Then join each clause-vertex to the vertices representing the literals in that clause. Call the resultant graph G_ϕ .

For example, if $\phi = (x \vee y \vee \bar{z}) \wedge (x \vee \bar{y} \vee w)$, then graph G_ϕ will look like this:



Claim: ϕ has a satisfying assignment if and only if G_ϕ has a dominating set of size k , where k is the number of variables used in ϕ . (In the example $k = 4$.)

Assume G_ϕ has a dominating set S of size k . Then, since one needs to dominate the unlabeled vertex in each variable-gadget, the set S has exactly one vertex from each variable-gadget, and each clause vertex is dominated by one of the literals. Thus one can create a satisfying assignment for ϕ by setting x to be true if $x \in S$ and x to be false if $\bar{x} \in S$. Conversely, if one has a satisfying assignment, then one can create a dominating set of G_ϕ by adding x to the set if x is true and $\bar{x}$ to the set if x is false.

13.3.2 Domination: Proof Two

Suppose instead that the VERTEXCOVER problem has already been proven to be NP-complete, where the input to that is a graph and a parameter and the question is whether the graph has a vertex cover of at most that size. Then another way to prove that DOMINATION is NP-complete is to reduce VERTEXCOVER to it.

Given a graph G , one can construct graph G' by, for each edge e , adding a new vertex w_e whose neighbors are the ends of e . The claim is that the domination number of G' equals the vertex cover number of G . Any dominating set D of G' must dominate each w_e . We can always choose D to not contain any of the w_e by replacing such a w_e by either of its neighbors. This means we can choose D

to be a vertex cover of G . Conversely, any vertex cover of G is immediately a dominating set of G' .

So the problem of calculating the domination number is at least as hard as the problem of calculating the vertex cover number. Or put in the terminology above, we reduce VERTEXCOVER to DOMINATION by mapping (G, k) to (G', k) .

13.3.3 Defective Coloring

Consider the problem of coloring a graph with two colors, say red and blue, such that every red vertex has no red neighbor, and every blue vertex has at most one blue neighbor. Call this the DEFECTIVE(0,1)-COLORING problem. As usual it is trivial to check that the problem is in NP: the certificate is the actual coloring.

THEOREM. *The DEFECTIVE(0,1)-COLORING problem is NP-complete.*

We will reduce from ONE-IN-THREE SAT. (The book by Garey and Johnson has a cornucopia of NP-complete problems.)

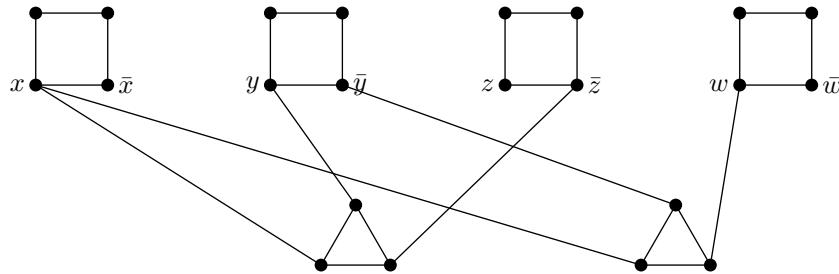
ONE-IN-THREE SAT

Input: Boolean formula ϕ with each clause having three literals

Question: Is there a truth assignment of the variables such that each clause in ϕ has exactly one true literal?

Now for the reduction: Consider the boolean formula ϕ . For each variable, say x , construct a 4-cycle with two consecutive vertices labeled x and $\bar{x}$. For each clause, construct a triangle, and add a matching between the triangle and the vertices corresponding to the literals in the clause. Call the resultant graph G_ϕ .

For example, if $\phi = (x \vee y \vee \bar{z}) \wedge (x \vee \bar{y} \vee w)$, then graph G_ϕ will look like this:



Claim: ϕ has a one-in-three satisfying assignment if and only if G_ϕ has a defective(0,1)-coloring.

Assume ϕ has a one-in-three satisfying assignment. Color each true literal blue and each false literal red. That is, if x is true in the assignment, then color x blue and $\bar{x}$ false; if x is false in the assignment, then color x red and $\bar{x}$ blue. Complete each 4-cycle alternating colors. Then in each clause triangle, color the vertex the opposite color of its literal. The clause has one blue and two red neighbors. The clause triangle is colored one red and two blues, and every vertex is okay.

Conversely, assume G_ϕ has a defective(0,1)-coloring. Then every variable 4-cycle must be colored alternating red and blue; thus the two literals in it receive opposite colors. And every clause triangle must have one vertex colored red and two colored blue. Further, the red vertex must have a blue neighbor in the vertex gadgets and both blues must have red neighbors in the vertex gadgets (since their quota of blue neighbors is used up). So exactly one of the literals in the clause is blue. Thus the set of blue literals corresponds to a one-in-three assignment of ϕ .

Thus we have shown that ONE-IN-THREE SAT reduces to DEFECTIVE(0,1)-COLORING problem, and so the latter is NP-complete.

13.3.4 Restricted 3-colorability

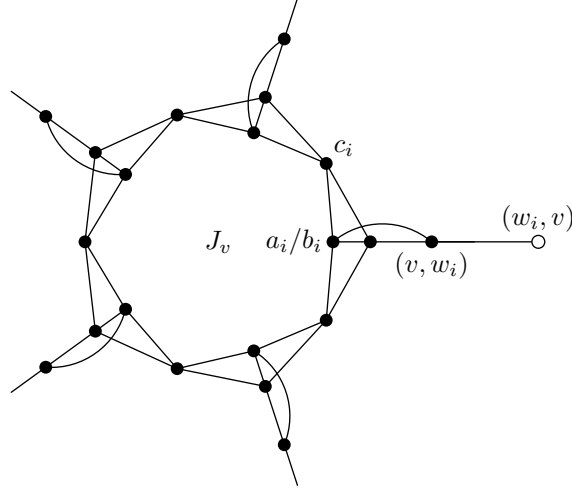
THEOREM. *The 3COLORABILITY problem (given graph G determine if $\chi(G) \leq 3$) is NP-complete even for graphs of maximum degree 4.*

This is a **restriction** of a problem known to be in NP, so it is automatically in NP itself.

We will reduce from 3COLORABILITY. Specifically, given graph G , we will construct graph H_G such that (a) H_G has maximum degree 4, and (b) H_G is 3-colorable if and only if G is 3-colorable.

Now for the reduction. For each vertex v in G , we construct a gadget J_v in H_G as follows. Say the neighbors of v in G are $w_1, w_2, \dots, w_d$. Then introduce vertices (v, w_1) through (v, w_d) . Then for $1 \leq i \leq d$, add vertices a_i, b_i, c_i and add edges $(v, w_i)a_i, (v, w_i)b_i, a_ib_i, a_ic_i, b_ic_i, a_{i+1}c_i$ and $b_{i+1}c_i$. Finally add edges joining

each (v, w_i) to the analogous vertex (w_i, v) in J_{w_i} . The result has maximum degree 4.



Now, suppose G is 3-colorable. Then color H_G as follows. For each vertex v in G , give its color to every vertex of the form (v, w_i) and every c_i in J_v . This can immediately be extended to a proper 3-coloring of H_G by giving the a_i, b_i the two colors other than that of v .

Conversely, suppose H_G is 3-colorable. Then, for each J_v , all the c_i receive the same color and thus so do all the vertices of the form (v, w_i) . We can thus color vertex v of G with the color of (v, w_i) , which is necessarily different to the color of each (w_i, v) .

Chapter 14: More Complexity Ideas

14.1 Approximation Algorithms

An *approximation algorithm* is one where the answer is guaranteed to be “close” to the actual. For example, consider vertex cover. Here is a factor-2 approximation algorithm, meaning it produces a vertex cover that is guaranteed to be at most twice the optimal size:

While there exists some edge, take both ends of the edge and remove.

Note that for every edge we remove, we need at least one end in every vertex cover.

It is known that: if $P \neq NP$, then there is no polynomial-time algorithm to approximate domination number within a constant factor. The best known algorithm has a logarithmic factor. Similarly, there exists a constant $\epsilon > 0$ such that one cannot approximate chromatic number to within a factor of n^ϵ .

For a long time, the best approximation result for 3-colorable graphs was: that one could find a coloring using $O(\sqrt{n})$ colors:

While there exists a vertex degree at least $\sqrt{n}$, 2-color its neighbors and remove them. Thereafter greedily color with $\sqrt{n} + 1$ colors.

14.2 Polynomial Hierarchy

One can also define NP as follows: a problem B is in NP if there is a polynomial-time decidable problem A of ordered pairs $\langle x, y \rangle$ such that

$$B = \{ x : \exists y : \langle x, y \rangle \in A \}.$$

The *polynomial hierarchy* allows for multiple quantifiers. For example, co-NP is the complement of sets in NP, or equivalently, we replace $\exists$ by $\forall$ in the above (and A by its complement).

The class Σ_2^P is the collection of all problems one gets from a polynomial-time decidable A of triples by defining

$$B = \{ x : \exists y \forall z : \langle x, y, z \rangle \in A \}.$$

(The P is for “polynomial-time”.) For example, a graph is said to be **2-clique colorable** if one can color the vertices with 2 colors such that no *maximal* clique is monochromatic. This problem is in Σ_2^P . (Indeed this problem is Σ_2^P complete.) The complement of Σ_2^P is Π_2^P . In this notation, NP is Σ_1^P . One can then define further classes by adding more (alternating) quantifiers. The union of these is called the *polynomial hierarchy*.

14.3 PSPACE

The set PSPACE is the set of all decision problems that can be decided in polynomial space. It can be shown that it includes all of the polynomial hierarchy. Graph-theoretic games are often PSPACE-complete. These are games where two players take turns, where the goal of one player is to create something and the goal of the other player is to avoid something. For example, a game where k colors are given and the vertices are numbered, and the player must properly color the next vertex.

14.4 #P

The class “sharp-P” or “number-P” is a set of counting problems. Surprisingly, counting the number of perfect matchings in a graph has the same complexity as counting the number of 3-colorings (both are #P-complete), even though the existence questions are not (or rather are believed not) the same complexity.

14.5 Fixed-parameter tractability

Some NP-complete problems have a parameter as part of the input. Sometimes fixing the parameter does not change the complexity. For example, the chromatic

number problem is NP-complete; and even if one restricts to asking about 3-colorability, the problem remains so. In contrast, the genus-of-a-graph problem is NP-complete. But for any fixed genus there is a fast algorithm that determines whether a graph embeds on that surface.

A parameter π is called ***fixed-parameter tractable*** if there is an algorithm for the decision problem $\pi(G) \leq k$ that runs in time $f(k)n^C$ for some constant C and function f . For example, the genus is such a parameter. On the other hand domination is believed not to be so.

Vertex cover is fixed-parameter tractable. Here is the algorithm. It uses ***kernelization***; meaning we trim the graph to a constant-size graph. To determine if the vertex cover number is at most k :

- (1) *If there is a vertex of degree more than k , then take v . (If we don't take v , then we need more than k vertices.) Repeat.*
- (2) *Throw away isolated vertices.*
- (3) *If the graph at this point (meaning $\Delta \leq k$) has more than k^2 edges, then it does not have a vertex cover of size k .*
- (4) *If not, we have a graph with at most k^2 edges and no isolates. It follows that the number of vertices is at most $2k^2$. And now, simply use exhaustive search. This takes time exponential in k but does not depend on the order of the original graph.*

Chapter 15: Minors and Algorithms

15.1 Minors and Well-quasi-orderings

A **quasi-ordering** is a reflexive transitive relation. A set X is **well-quasi-ordered** (WQO) by quasi-ordering $\leq$, if given any infinite subsequence in X there must exist two elements x_i, x_j with $i < j$ such that $x_i \leq x_j$. It can readily be shown that:

FACT. *A set is WQO iff it contains neither an infinite antichain (no two elements comparable) nor an infinite strictly decreasing sequence.*

A graph G is a **topological minor** of graph H if H contains a subgraph that is a subdivision of G . People also say that G is a homeomorphic subgraph of H .

THEOREM. *[Kruskal] The finite trees are WQO by the topological minor relation.*

That is, given any infinite sequence of trees, there is $i < j$ such that tree T_j contains a subdivision of tree T_i as a subgraph.

THEOREM. *[Robertson & Seymour] The finite graphs are WQO by the minor relation.*

As an immediate corollary we get:

THEOREM. *Every graph property closed under minors has a characterization by a finite list of forbidden minors.*

In particular, for every surface there is a finite list of forbidden minors. Actually, they proved this consequence as a stepping stone to the overall theorem.

For example, a **linkless embedding** of a graph is an embedding of the graph in 3 dimensions such that there are no two interlocking cycles. Having a linkless embedding is clearly closed under taking minors. So before the problem was not known to be decidable; now it is known to be in P, since there is a polynomial time algorithm to test for the existence of a fixed minor. But note that the algorithm has a huge hidden constant, and is nonconstructive.

15.2 Treewidth

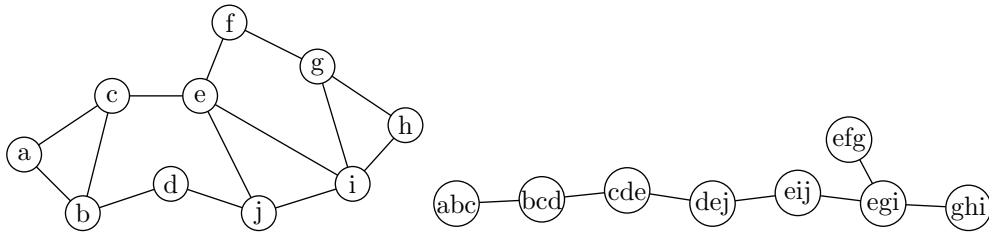
A **tree decomposition** of a graph G is a tree T and a mapping from the vertex set of T to the induced subgraphs of G such that:

- 1) every vertex of G is in V_t for some $t \in T$;
- 2) every edge of G has both ends in V_t for some $t \in T$;
- 3) if vertex t_2 lies on the t_1 - t_3 path in T , then $V_{t_1} \cap V_{t_3} \subseteq V_{t_2}$.

We call the sets V_t the **pieces** of G .

The width of a decomposition is 1 less than the maximum number of vertices in V_t . The **treewidth** of a graph is the minimum width of a tree decomposition.

Example 1. Here is a graph G and a tree decomposition T of width 2:



Example 2. There is always a trivial tree decomposition where the tree T has one vertex and its piece is the whole of G .

Example 3. If G is a tree, then T is the subdivision of G : If vertex t of T is a vertex of G , then V_t is just the vertex; if vertex t of T is an edge of G , then V_t is the edge and its ends.

Example 4. A cycle has treewidth 2. So a connected graph has treewidth 1 if and only if it is a tree.

A graph property is called **hereditary** if it is closed under taking subgraphs. Note that in the natural way, having treewidth at most k is hereditary. The graphs of treewidth k are also known as the **partial k -trees**, meaning subgraphs of k -trees. A **k -tree** is built by starting with K_{k+1} and at each stage adding a vertex and joining it to a k -clique. The graphs of treewidth at most 2 are also known as the **series-parallel** graphs, which are characterized as those graphs without K_4 as a minor.

THEOREM. (Robertson & Seymour) The graphs of treewidth at most k are WQO by minors.

Example 5. The $n \times n$ grid G_n has treewidth at least n .

THEOREM. (Robertson & Seymour) A graph of sufficiently large treewidth contains either K_m or the grid G_m as a minor.

15.3 Postorder Traversal Tree Algorithms

Many graph parameters can be calculated on trees by a postorder traversal algorithm. In such an algorithm, one picks any vertex as root and then processes all vertices such that each vertex is processed after its children. At each vertex a tuple of values is calculated using the values for the children. The overall algorithm runs in linear time provided processing a vertex runs in time proportional to the number of children.

The tuple of values calculated at a vertex v are usually the values of modified versions of the parameter applied to the subtree rooted at v , where the criterion is either relaxed or strengthened at vertex v . These criteria are determined by what states are possible if one restricts a valid set to that subtree. For example, in domination the states are that the root is in the set, the root is not in the set but dominated, or the root is not dominated.

15.3.1 Example: Independent Domination

For an example: consider the independent domination number of a tree. If we consider an independent dominating set S restricted to a subtree, there are three possibilities for the root of the subtree:

- (1) it is in S ,
- (2) it is not in S but dominated by one of its children,
- (3) it is not dominated within that subtree.

So let $f_i(T_v)$ be the minimum cardinality of a set of vertices in T_v such that the set S can be extended to an independent dominating set in the whole graph—that is, the set S is independent and dominates at least $V(T_v) - \{v\}$ —and the root v has property (i).

One can calculate these parameters as follows. If v is going to be in, then each of its children must be in state 2 or 3. Thus,

$$f_1(T_v) = 1 + \sum_c \min(f_2(T_c), f_3(T_c)).$$

If v is not in, then none of its children can be in state 3. Further, if v is not dominated, then none of its children can be in state 1. Thus

$$f_3(T_v) = \sum_c f_2(T_c).$$

If v is dominated, then at least one of its children is in state 1. So $f_2(T_v)$ is the minimum sum of $f_1(T_c)$ or $f_2(T_c)$ over its children, subject to the constraint that at least one of these must be f_1 . This evaluates to:

$$f_2(T_v) = \begin{cases} \sum_c \min(f_1(T_c), f_2(T_c)) & \text{if } f_1(T_c) \leq f_2(T_c) \text{ for some } c, \\ \sum_c f_2(T_c) + \min_c f_1(T_c) - f_2(T_c) & \text{otherwise.} \end{cases}$$

Finally, having done the traversal, one can calculate the actual independent domination number of the tree as $i(T) = \min(f_1(T_r), f_2(T_r))$ where r is the overall root.

Furthermore, to determine the actual minimum independent dominating set, one can work back down the tree, looking at each stage at which of the f_i was taken at the parent and calculating whether the vertex is in or out.

15.4 Treewidth and Algorithms

One can extend the postorder traversal idea to graphs with bounded treewidth. Given a tree decomposition T , we root the tree arbitrarily. Then for a vertex $t \in T$, define G_t as the subgraph of G induced by the pieces corresponding to t and its descendants.

Note that if t_1 and t_2 are children of t in T , then graphs $G_{t_1} - V_t$ and $G_{t_2} - V_t$ are disjoint and not joined by an edge. This follows by the property of tree decomposition.

Now, to get a dynamic programming algorithm, do a postorder traversal on T . If we remove one piece, then we split the graph into components and the only interaction is with the piece. So we calculate, for each vertex t of T and every possible interaction with V_t , the optimum of the parameter for G_t given its behavior on V_t . This is calculated from such values for its children. The algorithm takes time linear in the number of vertices of T , but exponential or worse in the treewidth.

For example: suppose we want to calculate the domination number. Then for G_t we calculate for every possible $R, S \subseteq V_t$ the minimum number of vertices in a set D that dominates at least $V(G_t) - R$, such that $D \cap V_t = S$.

Part III

Further Topics

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Chapter 16: Generalized Colorings of Planar Graphs

We saw that the fundamental result about coloring planar graphs is that the chromatic number is at most 4. Further, every planar graph has at most $3n - 6$ edges, where n is the order, and thus a vertex of degree at most 5. We consider here variations: replace planar graph by something more general or less general, and/or replace chromatic number by something stronger or weaker.

16.1 Extensions on Planarity

A graph is **1-planar** if it can be drawn such that every edge intersects at most one other edge.

THEOREM. (Borodin) *The chromatic number of a 1-planar graph is at most 6.*

A graph is **k -thick** if its edges can be partitioned into k planar graphs. Since a planar graph has average degree less than 6, a k -thick graph has average degree less than $6k$. In particular, a 2-thick graph has a vertex of degree at most 11, and hence (by induction) chromatic number at most 12. It is not known if this is best possible:

QUESTION. *What is the maximum chromatic number of a 2-thick graph?*

This is sometimes called the earth-and-moon problem.

16.2 Restrictions on Planarity

A graph is **outerplanar** if it can be drawn in the plane such that all vertices are on the outer cycle. A maximal outerplanar graph has exactly $2n - 4$ edges, and is formed by taking a hamilton cycle and triangulating the interior. An outerplanar graph can be characterized as having neither subdivision of K_4 nor a subdivision of $K_{2,3}$ as a subgraph.

THEOREM. *The chromatic number of an outerplanar graph is at most 3.*

In fact, the key to this is that every outerplanar graph has a vertex of degree at most 2. (Proof: induction.) We say that a graph is **k -dependent** if it and

every subgraph has a vertex of degree at most k . So a tree is 1-dependent, an outerplanar graph is 2-dependent, and a planar graph is 5-dependent.

Because of the characterization of Eulerian graphs, the term ***Eulerian*** is often used to mean every degree is even:

THEOREM. *The chromatic number of an Eulerian maximal planar graph is at most 3.*

(Proof: nice! Induction.) In fact there is a converse: if a planar graph is 3-colorable then it can be extended to an Eulerian planar graph.

A harder theorem is:

THEOREM. (*Grötzsch*) *A triangle-free planar graph is 3-colorable.*

A nice 5-color theorem is:

THEOREM. (*Borodin*) *A planar graph has a proper 5-coloring such that there is no 2-colored cycle.*

16.3 Two-Color Theorems

We noted that outerplanar graphs are 3-colorable. So what about coloring them with 2 colors?

It is not hard to show that one can partition the vertices of an outerplanar graph into two forests. (The minimum number of forests that one can partition the vertex set into is called the graph's (vertex) ***arboricity***.) But one can improve that:

THEOREM. *One can partition the vertices of an outerplanar graph into an independent set and a forest.*

PROOF. Induction. Remove vertex v of minimum degree, apply the inductive hypothesis, and then v can be added either to the independent set or to the forest. (Actually true for any 2-degenerate graph.) $\square$

A **linear forest** is a graph that is the vertex-disjoint union of paths. Several authors showed the following result. But there is a simpler proof.

THEOREM. *One can partition the vertices of an outerplanar graph into two linear forests.*

PROOF. Start at any vertex v . Let A be the vertices at even distance from v and B the vertices at odd distance from v . We claim that this is a suitable partition.

Note: we only need to show that the vertices at a specific distance from v form a linear forest. Let L be such a layer. Can check that a $K_{1,3}$ in L would mean a subdivided $K_{2,3}$ in the graph, which is not outerplanar. Also a cycle in L would mean a subdivided K_4 in the graph, which is not outerplanar. $\square$

(Woodall extended these ideas, and looked at choosability.)

THEOREM. *One can partition the vertices of an planar graph into two outerplanar graphs.*

PROOF. Again partition the vertices by parity of distance from some vertex. $\square$

Note that there is a similar, but much harder, result for edge partitions.

16.4 Three-Coloring a Planar Graph

It is not hard to show that one can partition the vertices of a planar graph into three forests. (Induction.) But actually Stein proved:

THEOREM. *One can partition the vertices of an planar graph into two forests and an independent set.*

This was then strengthened by Borodin's 5-color theorem about acyclic coloring.

Cowen, Cowen, and Woodall showed that one can partition the vertices of a planar graph into three graphs of maximum degree at most two. This was extended independently by Poh and the author. The proof uses strong induction.

THEOREM. *One can partition the vertices of a planar graph into three linear forests.*

Given a coloring of the vertices, we define a **color component** as a component of the subgraph induced by any of the colors. So the statement of the theorem is equivalent to saying there is a 3-coloring where every color component is a path.

PROOF. We prove a stronger statement by induction: that if any two adjacent vertices u and v are designated, it is possible to color the graph G into three linear forests such that neither u nor v has a neighbor of their own color.

The base case is trivial; e.g. a graph with say two vertices. In general, if say vertex u has degree at most 2, we can remove it, apply the inductive hypothesis to what remains with designated vertices v and any of its neighbors, and then add back u giving it the color missing from its neighborhood. So we may assume both u and v have degree at least 3.

Proceed as follows. Subdivide the edge uv by adding a vertex x in the middle of the edge. Then add edges to make the graph maximal planar without re-adding the edge uv . Since this is a maximal planar graph, the neighborhood of vertex u contains a cycle, and such a cycle separates u from v . Out of all separating cycles, let C be one that is as short as possible. Note that C contains x . Further C does not have any chords, since otherwise one obtains a shorter separating cycle.

We can think of C drawn such that u is inside and v is outside. Let G_u be the graph obtained if one removes all vertices outside C and contracts all of C to a single vertex, call it c . Note that G_u is planar, and that vertices u and c are adjacent. So one can apply the inductive hypothesis to G_u to obtain a suitable 3-coloring where neither u nor c has a neighbor of their color. Similarly, let G_v be the graph obtained if one removes all vertices inside C and contracts all of C to a single vertex, called c . Apply induction hypothesis to it.

We can assume that in G_u vertex u gets color 1 and vertex c gets color 2, and in G_v vertex v gets color 3 and vertex c gets color 2. Then take this coloring and lift it back to G so that all of C gets color 2. Note that in this coloring of G vertices u and v have no neighbor of their color. Further, since c had no neighbor of its color, no vertex of color 2 is adjacent to C . And since x is not a real vertex, in G we have only the color component $C - x$, which is a path. $\square$

16.5 Colorings of Sparse Planar Graphs

There are many results on some generalized version of coloring on planar graphs with large girth. A common technique is to show that planar graphs with high girth are sparse, and that the coloring result is true for all sparse graphs. The former uses Euler's formula, like we've done before:

FACT. *If a planar graph has girth at least g , then the average degree is less than $2g/(g-2)$.*

PROOF. Recall that $V - E + F = 2$. Further, the boundary of every face must have length at least g , but as before each edge is on the boundary of (at most) two faces, and thus we have $2E \geq gF$. Putting these two formulas together, some arithmetic yields $2E/V < 2g/(g-2)$. $\square$

The **maximum average degree** of a graph G , denoted $mad(G)$, is the maximum over all induced subgraphs of the average degree. Since subgraphs of planar graphs are planar, the above fact also holds for maximum average degree.

There is a general strategy to prove coloring results for sparse graphs. The first step is to identify some situations that allow induction. The second step is to show that when none of these situations occurs, there is a lower bound on the average degree. The latter is often proven by **discharging**. We give an example. Note that better results are known.

THEOREM. *If a graph G has $mad(G) < 5/2$, then one can 2-color the vertices such that every vertex has at most one neighbor of its color.*

PROOF. Three situations allow induction: (a) a vertex of degree at most 1; (b) two adjacent vertices of degree 2; (c) a degree-3 vertex all of whose neighbors are degree 2. In each case one removes the one/two/four vertices, applies the inductive hypothesis to the remaining graph H , gives each deleted vertex the color opposite to its neighbor in H , and in (c), gives the degree-3 vertex the minority color from its neighborhood.

The final step is to show that if a graph has none of the three situations described above, then its average degree is at least $5/2$. Start by giving each vertex a "charge" equal to its degree. At the beginning, the average charge

equals the average degree. Then the *discharging rule* is that each vertex of degree at least 3 sends $\frac{1}{4}$ charge to each degree-2 neighbor.

Every degree-2 vertex receives charge from both its neighbors, and so it ends up with charge exactly $5/2$. Every degree-3 vertex sends charge to at most two neighbors, and so it ends up with charge at least $5/2$. Every vertex with larger degree ends with at least $\frac{3}{4}$ of its original charge, and so ends up with charge at least 3. In other words, at the end every charge and hence the average charge is at least $5/2$. But this equals the average charge at the beginning. $\square$

Chapter 17: Colorings of Graphs on Surfaces

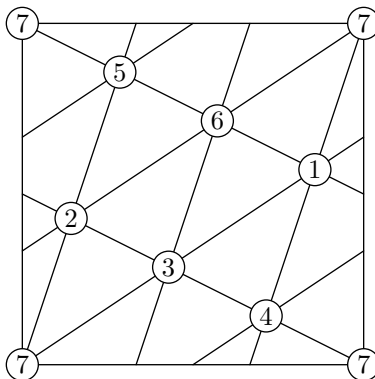
17.1 Genus and Orientable Surfaces

There are two types of surfaces: orientable and nonorientable. We consider here only the orientable type.

The plane and the sphere are equivalent for embedding purposes. In general one can take the sphere and add h handles. The resultant surface is called S_h and h is the **genus** of the surface. Equivalently the genus is the number of holes. The **torus** (or donut) has genus 1 and the pretzel has genus 2. The **genus** of a graph is the minimum genus of a surface needed to embed the graph. (Exists since worst-case one adds a handle for each edge.)

Now, it is important that there are two types of cycles on the surface. A **contractible** cycle can be continuously deformed or contracted to a single point. In the plane, all cycles are contractible, but on other surfaces they are not. If you slice through the torus on a noncontractible cycle, this will break the handle and you get a sphere; if you slice through the torus on a contractible cycle, you will get two pieces, a sphere and a torus.

For example, a torus can be obtained from a **rectangle** where the top and bottom are glued together, while the left and right are glued together. We say that the opposites sides are identified. Here is K_7 embedded on the torus, using this picture: the vertex at all four corners is the same vertex. In this example, cycle 136 is contractible, but cycle 123 is noncontractible.



There is an analogue of Euler's formula for a suitable definition of embedding. An embedding is a **2-cell** embedding if the boundary of each face is a contractible cycle.

THEOREM. *For a 2-cell embedding of a graph G on S_h , it holds that $V - E + F = 2 - 2h$ where V =vertices, E =edges, F =faces.*

PROOF. The proof is by induction on h . We already saw the result for $h = 0$.

So consider general S_h . Draw a noncontractible cycle C on S_h that does not go through any vertex of G . Since we have a 2-cell embedding, the cycle does not lie completely inside a face, and hence intersects some of the edges of G .

Now form a new graph G' by (a) adding vertices at every intersection between C and edges of G , and (b) adding edges embedded along C . The result is still a 2-cell embedding on the same surface.

Say there were k intersections between C and edges of G . Then G' has $V + k$ vertices, $E + 2k$ edges (k subdivisions and k new edges) and $F + k$ faces.

Now, cut S_h along the middle of C , thereby breaking the handle—called a capping operation. At the same time, duplicate each vertex and edge of C . The result is a graph G'' that has $V + 2k$ vertices, $E + 3k$ edges. Note that the cut exposes two new faces, bounded by the two copies of C . So the embedding has $F + k + 2$ faces, while it is on the surface with one less handle. It follows by the induction hypothesis that

$$(V + 2k) - (E + 3k) + (F + k + 2) = 2 - 2(h - 1).$$

And the result follows. $\square$

It follows that: $E \leq 3V + 6(h - 1)$.

17.2 Colorings

From Euler's formula, a graph on the torus has $E \leq 3V$ and so always has a vertex of degree at most 6. Hence

THEOREM. *The chromatic number of a toroidal graph is at most 7.*

This is best possible since K_7 has genus 1. Actually, any graph on the torus that requires 7 colors must contain K_7 as a subgraph. (Proof: induction/Brooks.)

In general, Euler's formula says that for any fixed surface, the average degree of a graph is $6 + o(1)$. From this there is an immediate upper bound on the chromatic number, by taking out a vertex of minimum degree and inducting. This bound turns out to be best possible. The hard part is showing that the largest clique that Euler's formula allows actually embeds on the surface.

There are also generalized coloring results for surfaces. For example, for the torus it is known that can partition the vertices of a toroidal graph into 5 sets such that each color component is either K_1 or K_2 . That is, each vertex has at most one neighbor of the same color. It is conjectured that the 5 can be replaced by 4. Another conjecture is that there always exists a triple of vertices such that when they are removed what remains is 4-colorable. In other words, yes seven colors are needed for the torus, but three of the colors need only be used once. (Note that this implies the 4-color theorem.)

Chapter 18: The Probabilistic Method and Random Graph

18.1 The Original Result

The original use of the probabilistic method was to provide a lower bound for Ramsey numbers, namely that $R(m, m) > 2^{m/2}$.

ERDŐS. *There exists a graph G on $N = 2^{m/2}$ vertices that has neither an independent set nor a clique of size m .*

PROOF. Start with N vertices. For each pair of vertices, toss an unbiased independent coin to see whether they are joined by an edge or not. The result is a graph G_N . We want to calculate the probability that G_N has the desired property.

The probability that any fixed set A of m vertices forms a clique is $\frac{1}{2^{\binom{m}{2}}}$, since every edge must be present. The probability that A forms an independent set is the same.

Thus, the probability that some subset of size m forms a clique or independent set, is at most $\binom{N}{m}$ times this quantity. That is,

$$\Pr(\exists \text{ bad } A) \leq (\# \text{ possible } A) \times \Pr(\text{specific } A \text{ is bad}) = \binom{N}{m} \times \frac{2}{2^{\binom{m}{2}}}.$$

Now, by a bit of arithmetic,

$$\binom{N}{m} \times \frac{2}{2^{\binom{m}{2}}} < \frac{2N^m}{m! 2^{m(m-1)/2}} < \frac{N^m}{2^{m^2/2}} = \left(\frac{N}{2^{m/2}}\right)^m = 1.$$

So,

$$\Pr(\exists \text{ bad } A) < 1.$$

Or put the other way around, the probability that there is no bad A is nonzero: one of these random graphs has neither an independent set nor a clique of size m . Thus there exists a graph on N vertices with the desired property. $\square$

Note that a probabilistic proof like this can also be thought of as a counting argument. We count all the graphs on N vertices. Then we count all the graphs on N vertices with a large clique or large independent set, and find that they cannot account for all the graphs.

18.2 Linearity of Expectation

Suppose set X is constructed by some random process. Suppose each element x is in X with probability π_x . Then, even though whether one x is in X might depend heavily on whether another x' is in X or not, by linearity of expectation the expected size of X is $\sum_x \pi_x$.

As an application:

THEOREM. *For any graph G there exists a partition of the vertices of G into two sets L and R such that at least $1/2$ the edges go between L and R .*

PROOF. Construct L and R at random by putting each vertex independently in L with probability $1/2$. The probability that a given edge goes between L and R is $1/2$ (Why?). So the expected number of edges between L and R is half the edges. Thus, there exists a partition where the number of edges is at most half the total and one where the number of edges is at least half the total. $\square$

Actually, it is not hard to explicitly construct such L and R .

THEOREM. *The domination number of an r -regular graph of order n is at most*

$$\gamma \leq \frac{1 + \ln r}{r} n.$$

PROOF. Constructing a set Y at random by taking each vertex independently with probability $p = \ln r/r$. Then create dominating set D by taking Y and adding every vertex that is not dominated by Y .

The probability a vertex v is in D but not Y is the probability that no vertex in $N[v]$ is in Y . This equals $(1 - p)^{1+r} \approx 1/r$. Hence the expected size of D is the value in the theorem. $\square$

Alon showed that there are graphs where the domination number is near this value. The proof also uses probabilistic methods.

18.3 Turán's Theorem

Recall:

THEOREM. The maximum number of edges in a graph on n vertices without a k -clique is $n^2(k-2)/(2(k-1))$.

Rephrased it reads:

THEOREM. If G is a graph with n vertices and q edges, then it has a clique of size at least $n^2/(n^2 - 2q)$.

PROOF. We start by showing a more general statement:

$$\text{clique-size} \geq \sum_{v \in V} \frac{1}{n - \deg v}.$$

Choose a random ordering π of the vertices. For a given ordering, let C_π be the set of all x such that x is adjacent to every vertex before it in the ordering π . This is a clique, as every vertex added to C_π is adjacent to all the earlier vertices of C_π .

Let us compute the expected size of C_π . Consider a vertex v and the set Z_v consisting of v and its non-neighbors. The probability p_v that $v \in C_\pi$ equals the probability that, out of all the vertices in Z_v , vertex v appears first in π . Hence this probability $p_v = 1/|Z_v|$.

By linearity of expectation it follows that the expected size of C_π equals $\sum 1/(n - \deg v)$. And there must be an actual clique with size at least the average. This proves the above claim.

Now, by calculus it follows that $\sum 1/(n - \deg v)$ for fixed q is minimized when the degrees are equal. So this sum is at least $n \times 1/(n - 2q/n)$. Hence there is a clique of this size. $\square$

18.4 Sperner's Theorem

We say that two sets are *incomparable* if neither is a subset of the other. A collection $\mathcal{S}$ of sets is called an antichain or a *Sperner family* if every pair of sets in $\mathcal{S}$ are incomparable. One can use probabilistic methods to find an upper bound on the size of a Sperner family:

THEOREM. Let $\mathcal{S}$ be a Sperner family of subsets of $\{1, \dots, n\}$. Then $|\mathcal{S}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

PROOF. We define a **chain** of sets as a sequence of subsets such that the next subset is obtained from the previous subset by adding one element.

Construct a chain $\mathcal{C}$ starting at $\emptyset$ and ending at $\{1, \dots, n\}$ at random. This means that at each stage we pick the next element to add to the set uniformly at random.

Let E denote the expected value of $|\mathcal{C} \cap \mathcal{S}|$. Then $E \leq 1$, because $|\mathcal{C} \cap \mathcal{S}| \leq 1$ for any chain $\mathcal{C}$. On the other hand, the probability that any particular set S of cardinality s is in $\mathcal{C}$ is $1/\binom{n}{s}$. (Why?) In particular, this probability is always at least $1/\binom{n}{\lfloor n/2 \rfloor}$. So

$$1 \geq E \geq |\mathcal{S}| / \binom{n}{\lfloor n/2 \rfloor},$$

which gives the bound. $\square$

Actually, the above proof can be used to prove the LYM inequality that extends Sperner's theorem.

18.5 Properties of the Random Graph

Let $G(n)$ be the graph on n vertices with edges present independently at random with probability $1/2$. We say a property holds **almost surely** if its probability of holding goes to 1 as n goes to infinity.

THEOREM. Almost surely $G(n)$ has diameter 2.

PROOF. Well, there is a vanishingly tiny probability of diameter 1. For vertices x and y , the probability that x and y have z as a common neighbor is $1/4$. The probability they have no common neighbor is therefore $(3/4)^{n-2}$. Thus the probability that there exists a pair not at distance two is at most n^2 times this probability, which is vanishing small. $\square$

THEOREM. The independence number of $G(n)$ is almost surely $2 \log_2 n + O(1)$.

Go back to the proof earlier. We argued that if $n = 2^{-k/2}$, then the probability that $G(n)$ does not have an independent set of size k was nonzero. That is, if $k = 2 \log_2 n$, then the probability that the independence number is less than k is nonzero. However, to show that it tends to 1 takes more work, in particular, bounds from probability theory such as Janson's inequality.

Bollobás showed:

THEOREM. *The chromatic number of $G(n)$ is almost surely $n/(2 \log_2 n)$.*

Note that the result on independence number gives the lower bound on the chromatic number, but it does not give an upper bound. Because, if we try to remove a maximum independent set and recurse, what remains is not a random graph.

Other facts that hold almost surely:

- the minimum degree is $n/2 - \sqrt{(n \log n)/2} (1 + o(1))$ with unique vertex of minimum degree.
- the connectivity equals the minimum degree.
- there is a perfect matching (up to one vertex) and indeed a hamilton cycle.
- there is no nontrivial automorphism.

There is a cute result about the infinite random graph:

THEOREM. *All random graphs with a countably infinite set of vertices are isomorphic.*

PROOF. Let $v_1, v_2, \dots$ be any ordering of the vertices of G_1 . Construct a bijection as follows. Pick a vertex w_0 in G_2 . Pick a neighbor or non-neighbor w_1 in G_2 depending on whether v_0 and v_1 are adjacent.

In general, let w_i be any vertex in G_2 such that bijection $v_j \rightarrow w_j$ for $1 \leq j \leq i$ is an isomorphism. Always exists because $1/2^{i-1}$ vertices in G_2 are candidates.

□

Other models of random graphs include random regular graphs and power law graphs.

18.6 Evolution of the Random Graph

One can generalize $G(n)$ to $G(n, p)$, which is the random graph on n vertices where every edge is present independently at random with probability p . It is useful to think of starting with $p = 0$ and steadily increasing p . It turns out that there is a “phase change” around $p = 1/n$.

If $p \gg 1/n$, then almost surely the graph is connected, and indeed hamiltonian, and many of the properties that we know for $G(n, 1/2)$ hold. Actually, one doesn’t have to go much above $1/n$. For example, the threshold for no vertex being isolated is around $p = \ln n/n$, and similarly for connectivity.

In contrast, if $p \ll 1/n$, then almost surely all components are trees, all of size at most $O(\log n)$. Cycles only appear when p is around $1/n$. Indeed, it turns out that the threshold for the appearance of a subgraph H is determined by its densest subgraph. Let $m(H)$ be the *maximum average degree* of a subgraph of H : that is, the maximum ratio $2s/r$ taken over all subgraphs of H on r vertices and s edges. Then the threshold is $p = n^{-2/m(H)}$.

Hence at $p = 1/n$, only unicyclic components are added. However, by $p = 2/n$ it turns out that almost surely there is one giant component that contains almost all the vertices. So the phase change happens at exactly $1/n$. Indeed, it can be proven that many of the thresholds are very abrupt.

Chapter 19: Matroids

The concept of matroids can be used to prove results in extremal set theory and graph theory, though they are also of interest in their own right. The following draws on the presentation in B. Bollobás, *Combinatorics*.

19.1 Three Theorems

Here are three famous results that can be proven using matroids. We saw Hall's theorem as a characterization for matching in bipartite graphs. Here is another formulation. Given a collection of sets, a ***system of distinct representatives*** is a distinct element out of each set. An obvious necessary condition is that for any sub-collection of k sets, the union of that sub-collection has at least k elements. Hall showed that this condition is sufficient:

THEOREM. [Hall] *A collection $\mathcal{C}$ of sets has a system of distinct representatives iff for all sub-collections $\mathcal{C}'$ sets it holds that*

$$\left| \bigcup \mathcal{C}' \right| \geq |\mathcal{C}'|.$$

A related problem is where the universe is some vector space and the sets then sets of vectors. The task here is to find for each set a representative such that the group of representatives is linearly independent. Again there is an obvious necessary condition: for any sub-collection of k sets, the rank of the union must be at least k . This condition is sufficient:

THEOREM. [Rado] *A collection $\mathcal{C}$ of sets of vectors has a system of distinct independent representatives iff for all sub-collections $\mathcal{C}'$ it holds that*

$$\text{rank} \left(\bigcup \mathcal{C}' \right) \geq |\mathcal{C}'|.$$

For a different type of result, consider a graph and a palette of k colors. The task is to color each edge such that the resultant coloring does not have a monochromatic cycle. Again there is an obvious necessary condition. The maximum number of edges that can be colored with one particular color is $n - 1$, where n is the order. And indeed this holds true for all subgraphs H . Surprisingly this is sufficient:

THEOREM. [Nash-Williams] *For any k and graph, one can assign each edge a color from $\{1, \dots, k\}$ such that the resultant coloring does not have a monochromatic cycle iff for all subgraphs H it holds that*

$$k \cdot (\#vertices(H) - 1) \geq \#edges(H).$$

19.2 Matroid Axioms

Matroids was first defined as a link between vectors and graphs as a way of capturing the combinatorial essence of linear independence. They can be defined in many ways. Here is one. A finite **matroid** has a universe X and a nonempty collection $\mathcal{I}$ of subsets of X , known as the **independent sets**, that obey the following two axioms:

1. $\mathcal{I}$ is closed under subsets. Meaning, $K \in \mathcal{I}$, $J \subseteq K$ implies $J \in \mathcal{I}$.
2. If $J, K \in \mathcal{I}$ with $|K| > |J|$, then there is an $a \in K$ such that $J \cup \{a\} \in \mathcal{I}$.

These axioms are motivated by linear algebra. The first says that a subset of an independent set is independent, and the second is based on the idea that every independent set is extendable to a basis.

19.3 Matroid Examples

Here are some examples.

- sets of vectors. The two properties are well-known.
- sets: where every subset is considered independent.
- a **graphical matroid**: the universe is the set of edges of a graph and a collection of edges is defined to be independent if it does not contain a cycle. (Verifying the axioms is left as an exercise.)

Many of the concepts from linear independence carry over. For example, one can define a **base** (or basis) as a maximal independent set. And one can define a matroid in terms of a **rank function** (mirroring linear algebra where the rank is the size of the largest independent subset). The axioms for a rank function are:

1. It obeys $0 \leq \text{rank}(S) \leq |S|$.
2. If $S \subseteq T$ then $\text{rank}(S) \leq \text{rank}(T)$.
3. It is **submodular**: $\text{rank}(S \cup T) + \text{rank}(S \cap T) \leq \text{rank}(S) + \text{rank}(T)$

Because of the results where a trivial necessary condition turns out to be sufficient, one often checks for a “matroidal property” when investigating. Unfortunately such cases are rare. Matroids are now more studied for their own worth. There are special classes of matroids (e.g. binary matroids) and standard operations to build new matroids (e.g. deletion, minors, and duals).

19.4 Matroid Theorems

To prove our main matroid theorem we need the following lemma about refining the independent sets. We say that a set A **spans** B if $\text{rank}(A) = \text{rank}(A \cup B)$.

LEMMA. Consider a matroid with independent sets $\mathcal{I}$. For fixed set B , define $\mathcal{J}$ to be those sets in $\mathcal{I}$ that do not span B . Then $\mathcal{J}$ forms a matroid too.

PROOF. (1) *Closure under subsets*:

Assume $J \subset K \in \mathcal{J}$. Then K is independent and does not span B , which implies $K \cup \{a\}$ independent for some $a \in B$, which implies $J \cup \{a\}$ is independent. This implies J is independent and does not span B . Thus $J \in \mathcal{J}$.

(2) *Extensibility*:

Assume $J, K \in \mathcal{J}$ with $|K| > |J|$. Then both J and K are independent and neither spans B . The latter implies $K \cup \{a\}$ is independent for some $a \in B$. There are two cases.

Case 1: If $J \cup \{a\}$ is independent, then since $|J \cup \{a\}| < |K \cup \{a\}|$, there exists $k \in K$ such that $J \cup \{a, k\}$ is independent. But this implies that $J \cup \{k\}$ is independent and does not span B .

Case 2: If $J \cup \{a\}$ is dependent, then because J does not span B , we have $J \cup \{b\}$ independent for some $b \in B$. Again there exists $k \in K \cup \{a\}$ such that $J \cup \{b, k\}$ independent. This implies $k \neq a$, and so $J \cup \{k\}$ is independent and does not span B . $\square$

We next prove the independent representatives theorem:

THEOREM. [Rado] *For any matroid, a collection $\mathcal{C}$ of sets (from the universe) has a system of independent representatives iff for sub-collections $\mathcal{C}'$ it holds that*

$$\text{rank}(\bigcup \mathcal{C}') \geq |\mathcal{C}'|.$$

PROOF. We proceed by induction on the number k of sets in $\mathcal{C}$. The result is trivial if $k = 0$. Let $S_1, \dots, S_k$ be the sets in $\mathcal{C}$. Let $\mathcal{J}$ be the independent sets defined by reclassifying as dependent any (independent) set that spans S_k . We saw above that this forms a matroid, say M' .

Further, we claim the revised system obeys the hypothesis of the theorem. For, let $Z = \bigcup \mathcal{C}'$ where $\mathcal{C}'$ is a sub-collection of $S_1, \dots, S_{k-1}$. If Z does not span S_k , then its rank is unchanged. So,

$$\text{rank}'(Z) = \text{rank}(Z) \geq |\mathcal{C}'|.$$

If Z does span S_k , then originally $\text{rank}(Z) = \text{rank}(Z \cup S_k) \geq |\mathcal{C}'| + 1$. Since $\text{rank}(S_k) \geq 1$ (by hypothesis), there is a maximal independent set I in Z that contains exactly one element a of S_k . So $I - a$ is independent, does not span S_k , is contained in Z and has size $|\mathcal{C}'|$. That is,

$$\text{rank}'(Z) \geq \text{rank}(Z) - 1 \geq |\mathcal{C}'|.$$

Hence the revised system has a system T of representatives that are independent in M' and hence independent in the original matroid. But by the definition of independence in M' , T does not span S_k ; so one can add $a \in S_k$ to T while retaining independence (in the original matroid), which completes the inductive step. $\square$

The covering theorem:

THEOREM. [Edmonds & Fulkerson] *For any matroid and positive integer k , the universe X can be partitioned into k independent sets iff for all subsets A of X it holds that*

$$k \cdot \text{rank}(A) \geq |A|.$$

PROOF. Take k copies of the universe X and color each copy with a different color. Construct matroid $\mathcal{S}'$ with this universe, where a set is independent if its restriction to each color is an independent set. Then construct collection $\mathcal{C}$ with each set consisting of the k colored copies of one element of X . This collection obeys the conditions of the previous theorem and so there is a system T of independent representatives. Note that a system of representatives is an assignment of colors. Further, the system being independent means that T restricted to each color is independent. That is, we have partitioned X into k independent sets. $\square$

Chapter 20: Geometric Graphs

We consider some several questions about general graphs drawn in the plane. When we restrict to straight-line segments for edges, these are sometimes called geometric graphs.

20.1 Crossing Number

We saw earlier that only relatively sparse graphs are planar. The question of genus is then an extension of planar. Another direction is to count the number of times two edges cross. The smallest value of this is called the *crossing number* of the graph. For example, it is not hard to show that the crossing number of both K_5 and $K_{3,3}$ is 1: if one removes one edge the result is planar, and one can add the missing edge and have it only cross one other edge.

The lower bound can be generalized to say that the crossing number is at least $E - (3V - 6)$, since each edge beyond $3V - 6$ must cross some edge. For example, it follows that K_6 has crossing number at least 3, and it can be drawn as such. But for dense graphs this bound is weak, since most edges will cross several edges. The crossing number of the complete graph is unresolved. It is known that:

THEOREM. *The crossing number of the complete graph lies between $\frac{1}{80}n^4$ and $\frac{1}{64}n^4 + O(n^3)$.*

20.2 Geometric Graphs

One can now consider graph questions as they relate to the geometry. For example, one might ask for the minimum number of edges that ensure one can find two edges that do not intersect. This was resolved by Erdős and others. We assume no three vertices are colinear.

THEOREM. *If a geometric graph with more edges than vertices, then there exists two disjoint edges that do not cross.*

PROOF. If there is a vertex of degree 1, then remove the vertex and its edge, and apply induction. So assume the minimum degree is at least 2. Let v be a vertex with degree more than 2, say with neighbors w_1, w_2, w_3 . If w_1, w_2, w_3 all lie in some half-plane created by a line through v , then let w_2 be the “middle” neighbor looking from v . Otherwise let w_2 be any neighbor. Then the second edge incident with w_2 cannot intersect both vw_1 and vw_3 . This gives us the desired pair of edges. $\square$

Then one can try to improve this to a set of k edges that form a matching in the graph and are non-intersecting. After several authors provided results for small k , Pach and Törőcsik used Dilworth’s theorem to show that a linear number of edges suffice.

Another direction is to consider that we have all edges, that is a drawing of the complete graph, and then the question is what size of non-intersecting or pairwise-intersecting matching is guaranteed. As before we require that no three vertices are colinear.

The answer to the first is perhaps surprising: one can find a (almost)-perfect matching none of whose edges intersect. The proof is to take the maximum matching with the shortest total length! If two of its edges cross, say ab intersects cd , then by the triangle inequality, one can replace this pair by either ac and bd , or by ad and bc , and so produce a shorter total length, a contradiction.

One can invert the second question and ask, how many vertices do we need to produce a crossing family of size k . For example, it is not hard to show that any drawing of K_5 must have a matching of size 2 whose edges intersect. This problem is related to the question of how many points does one need to ensure that there is a convex quadrilateral. In general, exponentially many vertices are needed.

Chapter 21: Computers in Graph Theory

We briefly review ways in which computer support might be used in graph theory research.

21.1 Data from Small Graphs

Researchers have compiled databases of all small graphs up through a certain order. For example, all graphs, all trees, all planar graphs, and all regular graphs. So one can try out a conjecture on small graphs or just get data that might suggest a pattern.

SPECIFIC EXAMPLE: The independent domination number of cubic graphs. We [2] ran a database search asking for the maximum independent domination number of a cubic graph of specific order. The biggest value relative to the order we found was $\frac{3}{8}$ (apart from tiny cases such as $K_{3,3}$). In fact there were two cubic graphs on 16 vertices that have $i = 6$, and two cubic graphs on 8 vertices that have $i = 3$. Using software to draw these graphs suggested a pattern where the order-16 graphs are made up of two order-8 pieces. So, this suggested a construction of graphs on $8m$ vertices. Now, to prove that the general graph does not have a smaller independent domination set, again a computer was harnessed to verify that dominating or almost dominating sets on each piece have suitable properties (though in theory this could be done by hand).

21.2 Exhaustive Search

Exhaustive search uses computers to try all possibilities in order to show that something does not exist.

A classic example is trying to find Ramsey numbers. That is, show that one can or cannot color the edges of K_n without having a red K_r or blue K_s .

SPECIFIC EXAMPLE: The bipartite Ramsey number $b(2, 2, 2)$. That is, how large can $K_{m,m}$ be and one still color the edges with three colors such that no monochromatic $K_{2,2}$. In particular, we [1] wanted to know if it was possible to 3-color the edges of $K_{10,10}$ to avoid monochromatic 4-cycle. Theory showed that

there is a unique bipartite graph H on 10 vertices and 34 edges without a 4-cycle. So one can hope to fit two copies of H in, and then check that what remains is also 4-cycle-free. And computer did this. Except, that the problem was already solved by Exoo twenty-five years before...

Much of the hard work in exhaustive search goes into exploiting symmetry to reduce the number of cases. One of the major tools is McKay's **nauty** program. Indeed, it was used in finding some of the databases mentioned above.

SPECIFIC EXAMPLE: In original Ramsey numbers, circulants were the extremal cases for small numbers, so perhaps they are in general. Thus, one might need to know the independence number of a circulant. Symmetry kicks in: since graph is vertex-transitive, one can start with any vertex v . Can assume the minimum displacement is from v to the next vertex clockwise, by rotation symmetry. If we are looking for an independent set of specific size, can assume that half are on the left, by mirror symmetry, and so on.

A more recent development is the idea of converting the problem to logic and then using a SAT solver.

21.3 Other Ideas

1. **Heuristic search.** This is used to try to find some graph or configuration, using search techniques developed in artificial intelligence. One example was for certain domination problems on grids, such as attacking all squares with queens.
2. **Specific Programming.** One of the first proofs to use computer programming was the proof of the 4-color theorem. Essentially that proof uses induction. Specifically, a reducible configuration is a subgraph that one can shrink in some way, apply induction, and recolor the whole thing. So the computer is used (a) to find these configurations, and then (b) to show that every planar graph must have one of them (unavoidable).
3. **Optimization.** Tools such as linear programming, integer programming, and computer algebra for calculus, are also useful.

21.4 References

- [1] Bipartite Ramsey numbers and Zaremkiewicz bounds, W. Goddard, M.A. Henning, and O. Oellermann. *Discrete Math.*, 219 (2000), 85–95.
- [2] On the independent domination number of regular graphs, W. Goddard, M.A. Henning, J. Lyle, J. Southey. *Ann. Comb.* 16 (2012), no. 4, 719–732.